An Explainable Deep Learning Framework for Conditional Domain Adaptation

PhD Thesis Proposal

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1 Abstract

Domain adaptation is a key approach to ensure that artificial intelligence models maintain reliable performance when facing distributional shifts between training (source) and testing (target) domains. These challenges are particularly acute in the analysis of neurophysiological signals, such as electroencephalography (EEG), where high inter-subject variability and non-stationarity often lead to critical failure modes. However, to achieve robust adaptation, the following challenges arise: (i) a loss of semantic coherence, which leads to failure modes like negative transfer and semantic collapse; (ii) high sensitivity to noisy pseudo-labels in the target domain, which degrades performance in unsupervised settings; and (iii) a lack of interpretability in the alignment mechanisms, which prevents validation of whether the model's decisions are based on relevant features.

In this context, this thesis proposal introduces an explainable deep learning framework for conditional domain adaptation. As a foundational contribution, we present Conditional Rényi α-Entropy Domain Adaptation (CREDA), a novel framework that integrates kernel-based conditional alignment with a differentiable, matrix-based formulation of Rényi's quadratic entropy. CREDA is structured around three core components: (i) a deep feature extractor for domain-invariant representations; (ii) an entropy-weighted strategy to enhance robustness by attenuating the influence of low-confidence pseudo-labels; and (iii) a class-conditional alignment loss to preserve semantic coherence and prevent class conflation.

The research plan outlined in this thesis will focus on the comprehensive validation of the CREDA framework across fundamentally different data modalities. Its performance will be evaluated on standard image classification benchmarks (Digits, ImageCLEF-DA, Office-31) and, crucially, on neurophysiological signals from a public motor imagery EEG dataset to address both cross-session and cross-subject classification tasks. Furthermore, a dedicated interpretability protocol using UMAP and Grad-CAM++ will be implemented to ensure the model's decisions are transparent and grounded in genuine class-relevant features—from object outlines in images to discriminative brain patterns in EEG. Collectively, this research aims to establish CREDA as a powerful and generalizable framework that advances the development of more reliable AI systems for critical real-world applications in both computer vision and neuroengineering.

2 Motivation

One of the fundamental challenges in the development of intelligent systems lies in their ability to maintain adequate performance when deployed in conditions that differ from those encountered during training [1]. In practice, training data often differ significantly in their statistical distribution from those observed in real-world conditions [2]. This discrepancy can compromise the generalization capacity of the model, even when the underlying task remains unchanged [3]. This phenomenon constitutes a critical source of performance degradation, evident even in models that have been correctly validated within their original domain [4]. The sensitivity of systems to such variations limits their scalability and robustness, highlighting that validation performance does not always translate into consistent results under real-world conditions [5]. These changes in data distribution are not isolated phenomena but occur naturally due to shifts in the environment, user groups, or data capture conditions [6], thereby creating a gap between the knowledge acquired in the source domain and the prevailing conditions in the target domain [7]. To address this issue, a set of strategies collectively referred to as Domain Adaptation has been developed, aiming to mitigate the effects of domain shift on model performance [8], as illustrated in Figure 1. Even in scenarios where domains share similar structures, small differences may lead to systematic prediction failures [9]. This vulnerability is exacerbated when models trained on labeled data in controlled contexts are later deployed in slightly altered conditions, leading to a considerable drop in performance [10]. Therefore, it becomes necessary to establish explicit mechanisms to address domain discrepancies, since automatic generalization has proven insufficient to ensure stable behavior [5]. In this regard, approaches that enable the reuse of

previously acquired knowledge without directly relying on source-domain data become especially relevant, particularly in contexts constrained by privacy, storage, or data availability [11]. Building more abstract and transferable internal representations not only reduces contextual dependency but also lays the groundwork for the development of more resilient, scalable, and sustainable models in real environments [12].

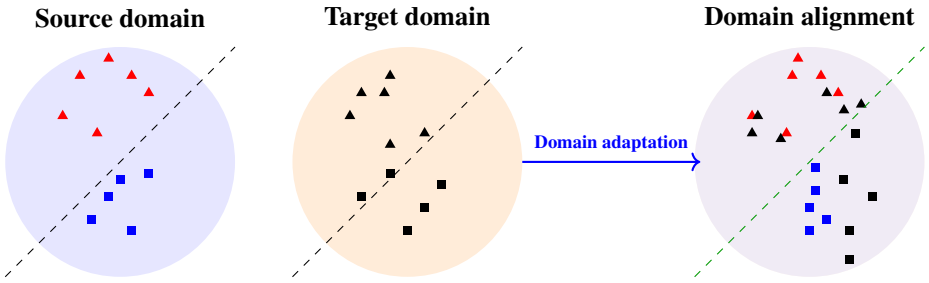


Figure 1: Domain alignment process. Visualization of domain adaptation from source to target domain.

In this context, Deep Learning (DL) architectures serve as a powerful engine for building sophisticated DA solutions, owing to their ability to learn complex and hierarchical representations from large volumes of data [13]. Their remarkable performance when training and testing data share the same distribution has led to their widespread adoption in diverse applications, from autonomous navigation to image-based species classification [14–18]. However, this success is predicated on a stable data distribution; the performance of these models is known to deteriorate significantly in the presence of domain shift, revealing a critical vulnerability in out-of-domain scenarios [19]. Therefore, it is crucial to understand that while DL provides the foundational components, its effective application in DA contexts requires explicitly training these architectures to generalize under new conditions [20, 21], thereby enabling the domain alignment process conceptualized in Figure 2.

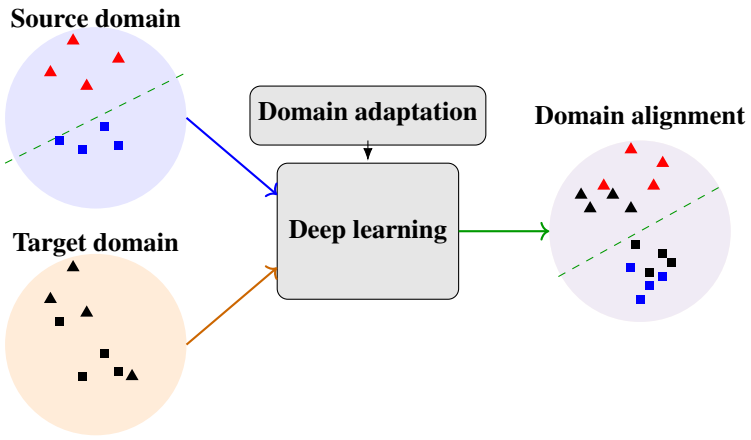
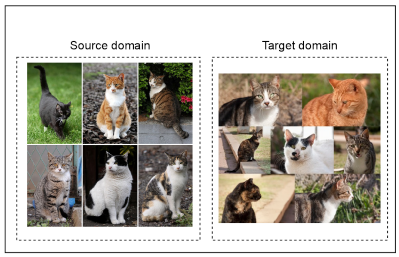


Figure 2: Domain alignment process. **Left**: Source and target domains. **Center**: Deep learning model performing feature transformation. **Right**: Domain-aligned representations. Dashed lines denote cluster separations and decision boundaries within the source and aligned domains.

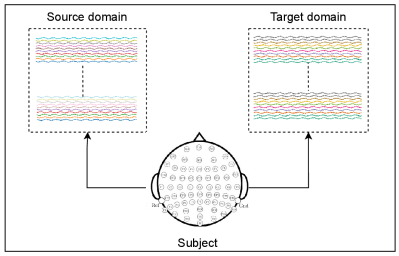
The need for advanced DA frameworks is particularly pressing in fields that handle highly variable and non-stationary data, making neurophysiological signal analysis a prime application area. Electroencephalography (EEG) data is notoriously challenging due to its low signal-to-noise ratio and susceptibility to multiple, co-occurring sources of variation. The Grupo de Control y Procesamiento Digital de Señales (GCPDS) at the Universidad Nacional de Colombia has a strong research trajectory in this area, having developed machine learning methodologies to address issues like inter-subject variability and dataset transfer in applications such as ADHD diagnosis and patient monitoring under anesthesia [22–24]. This body of work highlights the critical, real-world need for robust DA models that can generalize effectively in such complex settings and directly motivates the research proposed in this thesis. This research is conducted under the project "Alianza científica con enfoque comunitario para mitigar brechas de atención y manejo de trastornos mentales relacionados con impulsividad en Colombia (ACEMATE)-91908"—Project: "Sistema multimodal apoyado en juegos serios orientado a la evaluación e intervención neurocognitiva personalizada en trastornos de impulsividad asociados a TDAH como soporte a la intervención presencial y remota en entornos clínicos, educativos y comunitarios-790-2023", funded by Minciencias.

Building upon this context, the development of a novel DA framework is further justified by three specific and persistent forms of data variability that fundamentally limit existing models. A primary motivation is addressing the high dispersion of samples within a single semantic category. This phenomenon, well-documented in computer vision as intra-class variability (Figure 3a) where visual differences complicate classification [25], manifests in a more complex and dynamic form in EEG analysis as intra-subject variability (Figure 3b). Here, signals from the same individual change significantly across recording sessions due to physiological, emotional, or contextual factors [26], which severely impacts model stability and reliability [27]. This inher-

ent non-stationarity provides a strong impetus for designing DA strategies that can dynamically adapt to the evolving conditions of a single user, thus ensuring consistent performance over time [28].



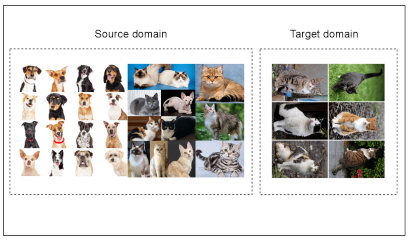
(a) Intra-class Variability.



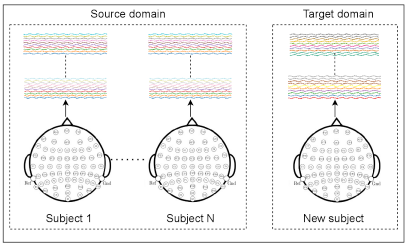
(b) Intra-subject Variability.

Figure 3: Intra-class and intra-subject variability. (**a**) Visual differences within a single image class. (**b**) EEG fluctuations across multiple sessions from the same individual.

A second, equally compelling, motivation arises from the significant differences that exist between distinct categories or individuals. In the image domain, this challenge is observed as inter-class similarity, where visual overlap between different classes can increase ambiguity and classification error (Figure 4a) [29]. This issue is magnified in the context of EEG, where inter-subject variability represents a critical barrier to generalizable models (Figure 4b). Each individual possesses a unique neurophysiological signature, or neural fingerprint, which prevents a model trained on one group of subjects from performing well on a new individual without extensive recalibration [30]. This heterogeneity, therefore, necessitates the development of DA methods capable of constructing shared, invariant representations that capture common neural patterns while being robust to individual-specific characteristics, thereby enabling effective generalization to new users [31, 32].



(a) Inter-class variability

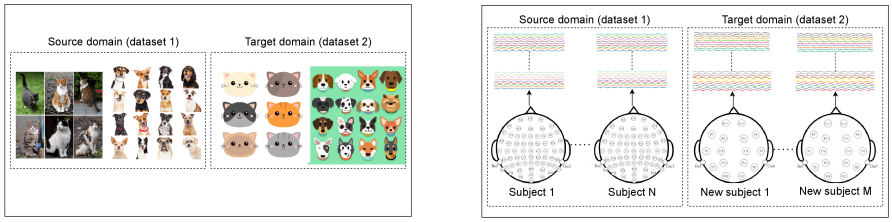


(b) Inter-subject variability

Figure 4: Inter-class and inter-subject variability. (**a**) Separation between distinct visual classes. (**b**) EEG signal differences across subjects.

Finally, the ambition to create truly robust and widely applicable models provides

the motivation to tackle cross-dataset variability, a widely acknowledged obstacle that emerges when attempting to transfer knowledge between datasets collected under disparate conditions [33, 34]. While this problem is common in many fields, such as image analysis (Figure 5a), it is particularly acute in EEG research (Figure 5b). Here, discrepancies in hardware (e.g., number of electrodes, amplifier types), experimental paradigms, and signal preprocessing pipelines create profound domain shifts that undermine model stability and reproducibility [35, 36]. Consequently, designing DA techniques that are resilient to these large-scale distribution shifts is a crucial research direction. Success in this area is essential for developing models that are not only high-performing but also reliable and efficient enough to be deployed across diverse clinical and research environments [37].



(a) Image datasets **(b)** EEG datasets

Figure 5: Cross-dataset variability. (**a**) Differences between image datasets. (**b**) Differences between EEG datasets.

In summary, the analysis of the aforementioned challenges—from the general limitations of DL in the face of domain variability to the specific and compounding issues of intra-subject, inter-subject, and cross-dataset adaptation in EEG—delineates a clear and compelling motivation for the development of novel and adaptive deep learning models.

3 Background

This framework must provide the tools to rigorously quantify distances between probability distributions and measure statistical dependence between variables in complex, high-dimensional spaces. This section lays the theoretical groundwork for such a framework, beginning with the fundamental geometry of Hilbert spaces, extending it to the functional domain of Reproducing Kernel Hilbert Spaces (RKHS).

3.1 Hilbert Space

A Hilbert space $\mathscr{H}$ is, in essence, a vector space over a field $\mathbb{K}$ (typically $\mathbb{R}$ or $\mathbb{C}$) endowed with an inner product $\langle \cdot, \cdot \rangle$. This product not only induces a norm $\| \cdot \|$ but also renders $\mathscr{H}$ a *complete* space with respect to the metric generated by that norm [38]. Completeness, by ensuring that every Cauchy sequence $\{\boldsymbol{x}_n\} \subset \mathscr{H}$ converges within

the space itself, constitutes the cornerstone of the analytical and spectral results that motivate the use of these spaces in applied mathematics and operator theory.

For all $\boldsymbol{x}, \boldsymbol{x}', \boldsymbol{z} \in \mathscr{H}$ and $a, b \in \mathbb{K}$, the inner product satisfies the axioms of linearity in the first argument, conjugate symmetry, and positive-definiteness:

$$\begin{aligned} \langle a\boldsymbol{x} + b\boldsymbol{x}', \boldsymbol{z} \rangle &= a\langle \boldsymbol{x}, \boldsymbol{z} \rangle + b\langle \boldsymbol{x}', \boldsymbol{z} \rangle, \\ \langle \boldsymbol{x}, \boldsymbol{x}' \rangle &= \overline{\langle \boldsymbol{x}', \boldsymbol{x} \rangle}, \\ \langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq 0 \quad &\text{and} \quad \langle \boldsymbol{x}, \boldsymbol{x} \rangle = 0 \iff \boldsymbol{x} = \boldsymbol{0}. \end{aligned}$$

From $\langle \cdot, \cdot \rangle$, the norm is defined as $\|\boldsymbol{x}\| = \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$, which in turn gives rise to the metric $d(\boldsymbol{x}, \boldsymbol{x}') = \|\boldsymbol{x} - \boldsymbol{x}'\|$. This metric-geometric structure allows for the treatment of continuity, convergence, and orthogonality within a unified functional language.

The squared distance between two vectors can be decomposed as:

$$\|\boldsymbol{x} - \boldsymbol{x}'\|^2 = \|\boldsymbol{x}\|^2 - 2\,\langle \boldsymbol{x}, \boldsymbol{x}' \rangle + \|\boldsymbol{x}'\|^2, \tag{1}$$

revealing that the cross-term $\langle \boldsymbol{x}, \boldsymbol{x}' \rangle$ acts as an analogue to the cosine in trigonometry. For non-zero vectors, this naturally leads to the definition of the angle $\theta_{\boldsymbol{x}\boldsymbol{x}'}$ by:

$$\cos\big(\theta_{\boldsymbol{x}\boldsymbol{x}'}\big) = \frac{\langle \boldsymbol{x}, \boldsymbol{x}' \rangle}{\|\boldsymbol{x}\|\,\|\boldsymbol{x}'\|}. \tag{2}$$

Orthogonality $\big(\langle \boldsymbol{x}, \boldsymbol{x}' \rangle = 0\big)$ reduces the distance to the mere sum of the squared norms, as derived from Equation (1).

Random Variables and Probability Spaces: To connect this geometry with stochastic analysis, consider a probability space $(\Omega, \mathscr{F}, \Pr)$ and a random variable $X : \Omega \to \mathbb{R}$ [39]. Here, Ω denotes the sample space, $\mathscr{F}$ the σ-algebra of events, and Pr a measure satisfying the Kolmogorov axioms.

Depending on the range $\chi = X(\Omega)$, two cases are distinguished:

- Discrete: Characterized by a probability mass function (PMF) $p(x) = \Pr(X = x)$, which satisfies $\sum_{x \in \chi} p(x) = 1$.
- Continuous: Characterized by a probability density function (PDF) $p(x)$ such that $\Pr(a \leq X \leq b) = \int_a^b p(x)\,dx$. The PDF satisfies $\int_{\mathbb{R}} p(x)\,dx = 1$.

Stochastic Hilbert Spaces: This transition to the probabilistic domain culminates in the space:

$$L^2(\Omega, \mathscr{F}, \Pr) = \big\{ X : \Omega \to \mathbb{R} \;\big|\; \mathbb{E}[X^2] < \infty \big\},$$

whose elements—square-integrable functions—once again form a Hilbert space when endowed with the inner product $\langle X, Y \rangle := \mathbb{E}[XY]$ and its associated norm $\|X\| = \sqrt{\mathbb{E}[X^2]}$. In this framework, variance, covariance, and, in general, statistical dependence are reinterpreted as distances, angles, and projections within a unified geometric framework, as follows:

- Expectation:
$$\mathbb{E}[X] = \langle X, 1 \rangle, \tag{3}$$
is the orthogonal projection of X onto the constant vector 1.

- Variance:
$$\mathrm{Var}(X) = \| X - \mathbb{E}[X] \|^2, \tag{4}$$
measures the dispersion as the norm of the centered version of X.

- Covariance:
$$\mathrm{Cov}(X,Y) = \langle X - \mathbb{E}[X], Y - \mathbb{E}[Y] \rangle, \tag{5}$$
extends the concept of the inner product to centered variables, where $Y : \Omega \to \mathbb{R}$ is another random variable.

- Correlation:
$$\rho(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}} = \cos(\theta_{XY}), \tag{6}$$
is interpreted as the cosine of the angle between the normalized versions of X and Y.

In particular, $\mathrm{Cov}(X,Y) = 0$ implies the orthogonality of the centered variables and, therefore, the absence of linear correlation.

Sample Estimation: In practice, the aforementioned parameters are estimated from independent and identically distributed samples $\boldsymbol{x} = \{x_m\}_{m=1}^M$ and $\boldsymbol{y} = \{y_m\}_{m=1}^M$ [40].

The population mean is approximated by:

$$\hat{\mu}_X = \frac{1}{M}\sum_{m=1}^{M} x_m; \quad \hat{\mu}_Y = \frac{1}{M}\sum_{m=1}^{M} y_m. \tag{7}$$

Analogously, the empirical Pearson correlation is expressed as:

$$\widehat{\rho}(X,Y) = \frac{\widehat{\mathrm{Cov}}(X,Y)}{\sqrt{\widehat{\mathrm{Var}}(X)}\sqrt{\widehat{\mathrm{Var}}(Y)}} = \frac{\frac{1}{M}\sum_{m=1}^{M}(x_m - \hat{\mu}_X)(y_m - \hat{\mu}_Y)}{\sqrt{\frac{1}{M}\sum_{m=1}^{M}(x_m - \hat{\mu}_X)^2}\sqrt{\frac{1}{M}\sum_{m=1}^{M}(y_m - \hat{\mu}_Y)^2}}. \tag{8}$$

Noting that subtracting the mean is equivalent to projecting each data vector onto the subspace orthogonal to $\mathbf{1}$, the above expression can be rewritten as:

$$\widehat{\rho}(X,Y) = \frac{\langle \boldsymbol{x} - \hat{\mu}_X \mathbf{1}, \boldsymbol{y} - \hat{\mu}_Y \mathbf{1} \rangle_2}{\| \boldsymbol{x} - \hat{\mu}_X \mathbf{1} \|_2 \, \| \boldsymbol{y} - \hat{\mu}_Y \mathbf{1} \|_2}. \tag{9}$$

Thus, the empirical correlation retains its geometric interpretation: it is the cosine of the angle between the centered data vectors, thereby bridging the gap between statistics and Hilbertian geometry.

3.2 Reproducing Kernel Hilbert Space (RKHS)

Extending the Hilbert space geometry from the previous section to the functional domain introduces the Reproducing Kernel Hilbert Space (RKHS). This space, denoted as $\mathscr{H}_\kappa$, consists of functions defined on a domain $\chi \subseteq \mathbb{R}^d$ and is determined by a symmetric, positive-definite kernel $\kappa : \chi \times \chi \to \mathbb{R}$, as illustrated in Figure 6.

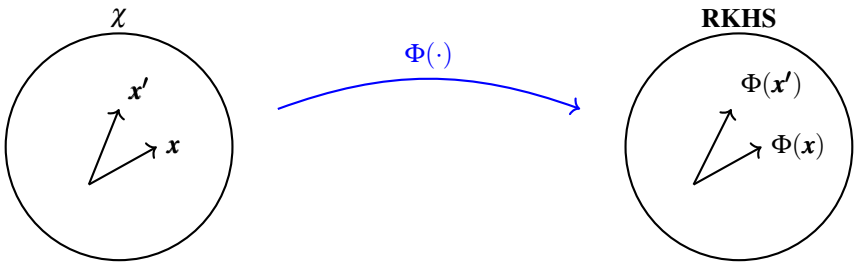


Figure 6: Mapping from the original domain χ to an RKHS via the feature map $\Phi(\cdot)$.

The *Moore–Aronszajn theorem* guarantees a bijective correspondence: every positive-definite kernel engenders a unique space $\mathscr{H}_\kappa$ in which the reproducing property holds [41]. For any $f \in \mathscr{H}_\kappa$ and $x \in \chi$:

$$f(x) \;=\; \langle f, \kappa(x,\cdot)\rangle_{\mathscr{H}_\kappa}, \tag{10}$$

where $\kappa(x,\cdot) \in \mathscr{H}_\kappa$ serves as the representer of point evaluation. Its existence is guaranteed by the Riesz representation theorem.

An explicit construction of the feature map $\Phi : \chi \to \mathscr{H}_\kappa$ is provided by *Mercer's theorem*. If the kernel admits the spectral expansion:

$$\kappa(x,x') \;=\; \sum_{n=1}^{\infty} \lambda_n \, \psi_n(x) \, \psi_n(x'), \tag{11}$$

with $\{\psi_n\}$ being an orthonormal basis in $L^2(\chi)$ and $\lambda_n \geq 0$, it suffices to define:

$$\Phi(x) = \big[\, \sqrt{\lambda_1}\,\psi_1(x),\, \sqrt{\lambda_2}\,\psi_2(x), \dots \big]^\top,$$

such that:

$$\kappa(x,x') \;=\; \langle \Phi(x), \Phi(x')\rangle_{\mathscr{H}_\kappa}. \tag{12}$$

From this identity, metric notions translate directly to the functional space:

$$\|\Phi(x)\|^2_{\mathscr{H}_\kappa} \;=\; \kappa(x,x), \tag{13}$$

$$\|\Phi(x) - \Phi(x')\|^2_{\mathscr{H}_\kappa} \;=\; \kappa(x,x) - 2\kappa(x,x') + \kappa(x',x'), \tag{14}$$

$$\cos\big(\theta_{\Phi(x),\Phi(x')}\big) \;=\; \frac{\kappa(x,x')}{\sqrt{\kappa(x,x)}\,\sqrt{\kappa(x',x')}}. \tag{15}$$

In this way, the algebraic-geometric machinery developed for Hilbert spaces is extended to the analysis of functions.

Stochastic Reproducing Kernel Space: The geometric characterization presented for RKHS can be extended to the stochastic domain by representing observations directly in $\mathscr{H}_\kappa$. Let $X : \Omega \to \chi$ (and analogously Y) be a random variable defined on the same probability space, and consider the feature map $\Phi : \chi \to \mathscr{H}_\kappa$. The compositions $\Phi(X)$ and $\Phi(Y)$ are then random elements taking values in a Hilbert space of—possibly—infinite dimension.

The set of such square-integrable variables is described by:

$$L^2(\Omega, \mathscr{F}, \Pr; \mathscr{H}_\kappa) = \Big\{ \mathbf{\Phi} : \Omega \to \mathscr{H}_\kappa \;\Big|\; \mathbb{E}\big[\|\mathbf{\Phi}(\omega)\|^2_{\mathscr{H}_\kappa}\big] < \infty \Big\},$$

a space that retains its Hilbertian structure when endowed with the inner product:

$$\langle \Phi(X), \Phi(Y) \rangle_{L^2(\mathscr{H}_\kappa)} := \mathbb{E}\big[\langle \Phi(X), \Phi(Y) \rangle_{\mathscr{H}_\kappa}\big]. \tag{16}$$

This construction allows for the translation of classical statistical moments into the new functional framework [42]:

- Kernel Mean Embedding: The analog of the expectation is defined as:

$$\mu_\kappa := \mathbb{E}[\Phi(X)] = \int_\chi \kappa(x, \cdot)\, d\Pr_X(x) = \int_\chi \kappa(x, \cdot)\, p(x)\, dx \;\in\; \mathscr{H}_\kappa. \tag{17}$$

- Functional Variance: The dispersion of X around μ_κ is given by:

$$\mathrm{Var}_\kappa(X) := \mathbb{E}\big[\|\Phi(X) - \mu_\kappa\|^2_{\mathscr{H}_\kappa}\big], \tag{18}$$

  this is a non-negative scalar value.

- Functional Covariance: The linear dependence between X and Y in the $\mathscr{H}_\kappa$ space is quantified via:

$$\mathrm{Cov}_\kappa(X, Y) := \mathbb{E}\big[\langle \Phi(X) - \mu_{\kappa,X}, \Phi(Y) - \mu_{\kappa,Y} \rangle_{\mathscr{H}_\kappa}\big]. \tag{19}$$

- Functional Correlation: Normalizing the covariance yields a dimensionless measure of dependence:

$$\rho_\kappa(X, Y) := \frac{\mathrm{Cov}_\kappa(X, Y)}{\sqrt{\mathrm{Var}_\kappa(X)}\,\sqrt{\mathrm{Var}_\kappa(Y)}}. \tag{20}$$

The geometric and stochastic tools developed thus far provide the foundation for defining kernelized measures of discrepancy and dependence, which are central to addressing the problem of domain shift. The first of these measures is the Maximum Mean Discrepancy (MMD). To define it, we consider a source distribution $X_s \sim \Pr_s$ and a target distribution $X_t \sim \Pr_t$, both defined on a common measurable space χ. Let their respective random variables be $X_s : \Omega \to \chi$ and $X_t : \Omega \to \chi$, which take values in their respective support domains $\chi_s \subseteq \chi$ and $\chi_t \subseteq \chi$. MMD then quantifies the distance that arises when these variables are embedded into the same space $\mathscr{H}_\kappa$ via the feature map $\Phi(X) := \kappa(X, \cdot)$ [43]; this process is illustrated in Figure 7.

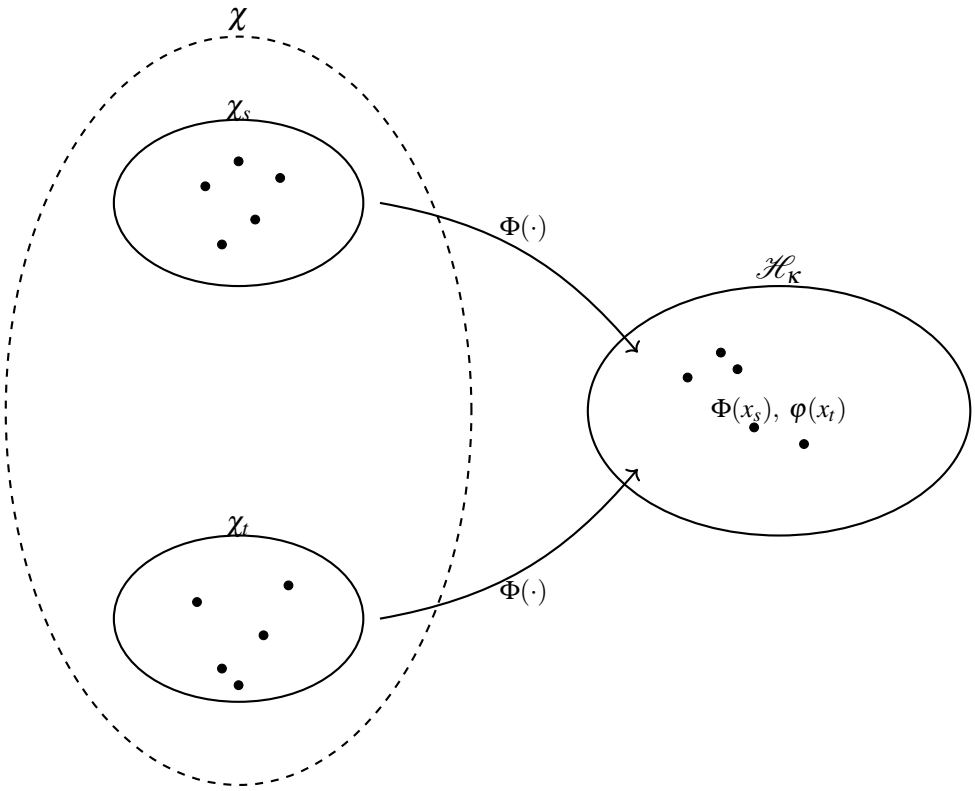


Figure 7: Illustration of the source X_s and target X_t sets projected onto a common RKHS $\mathscr{H}_\kappa$ via the feature map $\Phi : \chi \to \mathscr{H}_\kappa$ induced by the shared kernel κ.

We denote the squared discrepancy between the two kernelized distributions by $\mathrm{MMD}^2_\kappa(X_s, X_t)$:

$$\mathrm{MMD}^2_\kappa(X_s, X_t) := \left\| \mathbb{E}_{X_s}[\Phi(X_s)] - \mathbb{E}_{X_t}[\Phi(X_t)] \right\|^2_{\mathscr{H}_\kappa}. \tag{21}$$

Expanding the norm yields:

$$\mathrm{MMD}^2_\kappa(X_s, X_t) = \mathbb{E}_{X_s, X_s'}[\kappa(X_s, X_s')] + \mathbb{E}_{X_t, X_t'}[\kappa(X_t, X_t')] - 2\,\mathbb{E}_{X_s, X_t}[\kappa(X_s, X_t)], \tag{22}$$

whose empirical estimator, previously given in Eq. 7, takes the form:

$$\widehat{\mathrm{MMD}}^2_\kappa(X_s, X_t) = \frac{1}{M^2} \sum_{m,m'=1}^{M} \kappa(x_s^m, x_s^{m'}) + \frac{1}{M^2} \sum_{m,m'=1}^{M} \kappa(x_t^m, x_t^{m'}) - \frac{2}{M^2} \sum_{m,m'=1}^{M} \kappa(x_s^m, x_t^{m'}). \tag{23}$$

In matrix notation, with the Gram matrices $\mathbf{K}_{ss}, \mathbf{K}_{tt}, \mathbf{K}_{st}$:

$$\widehat{\mathrm{MMD}}^2_\kappa(X_s, X_t) = \frac{1}{M^2} \mathbf{1}_M^\top \mathbf{K}_{ss} \mathbf{1}_M + \frac{1}{M^2} \mathbf{1}_M^\top \mathbf{K}_{tt} \mathbf{1}_M - \frac{2}{M^2} \mathbf{1}_M^\top \mathbf{K}_{st} \mathbf{1}_M. \tag{24}$$

Whereas MMD measures the distance between the means of two distributions, a measure of dependence requires quantifying the relationship between their higher-order statistics. This is achieved by introducing the cross-covariance operator, a generalization of the covariance matrix to the setting of Hilbert spaces.

For variables X and Y with feature maps $\Phi : \chi \to \mathscr{H}_\kappa$ and $\Psi : \chi' \to \mathscr{H}_l$, the uncentered cross-covariance operator, $\mathscr{C}_{XY}$, is the expectation of the tensor product of their feature representations:

$$\mathscr{C}_{XY} := \mathbb{E}_{X,Y}[\Phi(X) \otimes \Psi(Y)]. \tag{25}$$

This object is a linear operator in the Hilbert-Schmidt space $\mathrm{HS}(\mathscr{H}_l, \mathscr{H}_\kappa)$, which can be conceptualized as the space of interactions between the two feature spaces (see Figure 8).

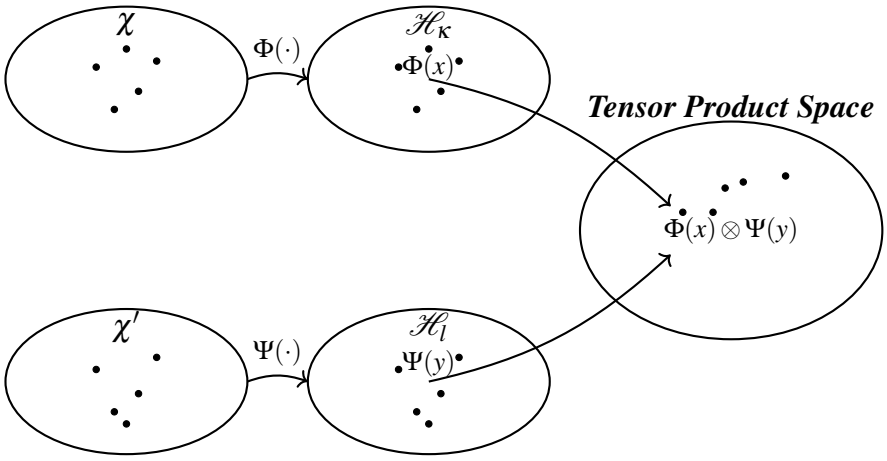


Figure 8: Mapping of the subspaces χ and χ' via the feature maps $\Phi(\cdot)$ and $\Psi(\cdot)$ onto their tensor product space.

Its fundamental property is that it allows for the computation of covariances between evaluation functionals: $\mathbb{E}_{X,Y}[f(X)g(Y)] = \langle f, \mathscr{C}_{XY} g \rangle_{\mathscr{H}_\kappa}$ for any $f \in \mathscr{H}_\kappa, g \in \mathscr{H}_l$ [44].

Statistical dependence is measured using the centered covariance operator, C_{XY}, which is obtained by subtracting the tensor product of the marginal means:

$$C_{XY} := \mathscr{C}_{XY} - \mu_\kappa \otimes \mu_l = \mathbb{E}_{X,Y}[(\Phi(X) - \mu_\kappa) \otimes (\Psi(Y) - \mu_l)]. \tag{26}$$

The second key measure, the Hilbert-Schmidt Independence Criterion (HSIC), is then defined as the squared norm of this centered operator. If X and Y are independent, C_{XY} is the null operator, and thus HSIC is zero [45].

$$\mathrm{HSIC}(X,Y) := \left\| C_{XY} \right\|_{\mathrm{HS}}^2 = \left\| \mathscr{C}_{XY} - \mu_\kappa \otimes \mu_l \right\|_{\mathrm{HS}}^2. \tag{27}$$

Expanding this norm reveals the population expression for HSIC in terms of kernels, which is analogous to the expansion for MMD:

$$\begin{aligned} \mathrm{HSIC}(X,Y) = \mathbb{E}_{X,X',Y,Y'}[\kappa(X,X')l(Y,Y')] + \mathbb{E}_{X,X'}[\kappa(X,X')]\mathbb{E}_{Y,Y'}[l(Y,Y')] \\ - 2\mathbb{E}_{X,Y}\left[\mathbb{E}_{X'}[\kappa(X,X')]\mathbb{E}_{Y'}[l(Y,Y')]\right]. \end{aligned} \tag{28}$$

This formulation provides a direct path to an empirical estimator (see Eq. 7). The empirical counterpart for a sample of size M expands as:

$$\begin{aligned}\widehat{\mathrm{HSIC}} = \frac{1}{M^2}\sum_{m,m'=1}^{M}\kappa(x_m,x_{m'})\,l(y_m,y_{m'}) + \frac{1}{M^4}\sum_{m,m',\hat{m},\tilde{m}=1}^{M}\kappa(x_m,x_{m'})\,l(y_{\hat{m}},y_{\tilde{m}}) \\ - \frac{2}{M^3}\sum_{m,m',\hat{m}=1}^{M}\kappa(x_m,x_{m'})\,l(y_{\hat{m}},y_{m'}).\end{aligned} \tag{29}$$

Using the centering matrix $\mathbf{H} = \mathbf{I}_M - \frac{1}{M}\mathbf{1}_M\mathbf{1}_M^\top$, a simpler (though biased) estimator is given in matrix form by:

$$\widehat{\mathrm{HSIC}}(X,Y) = \frac{1}{(M-1)^2}\mathrm{tr}(\mathbf{KHLH}) \tag{30}$$

where $\mathbf{K}$ and $\mathbf{L}$ are the Gram matrices corresponding to the samples of X and Y. These expressions complete the transition from kernelized moments to statistical tests for discrepancy and dependence, which operate exclusively in terms of kernels, thereby preserving the rich geometric structure of reproducing kernel Hilbert spaces.

4 Problem Statement

One of the main challenges in developing machine learning models is generalization to unseen domains, understood as the model's ability to maintain adequate performance on data whose distribution differs from that observed during training [46]. This issue is widely recognized for its impact on model robustness and applicability in real-world scenarios, where conditions may vary significantly and not all contexts can be anticipated [47]. Its relevance becomes evident in settings with high structural or semantic variability, which hinder performance consistency beyond the source domain [48]. Moreover, differences in data origin or nature can introduce subtle yet impactful distributional shifts, compromising model stability in new environments [49]. This phenomenon is particularly critical in tasks dependent on domain-sensitive features, such as language, where variations in style, genre, or topic can alter system behavior [50]. In dynamic environments where data evolve over time, the inability to generalize becomes a barrier to the reliable and effective deployment of automated solutions [51].

Beyond compromising robustness, lack of generalization to unseen domains entails structural consequences. First, models often overfit to the source domain, capturing spurious correlations or training-specific biases that limit their extrapolation capacity [52]. This leads to latent representations that are inadequate or degraded when exposed to new distributions [53]. Furthermore, theoretical guarantees for proper generalization usually require a sufficiently large number of source domains; in settings with few domains, these assumptions break down [48]. This situation is worsened by the presence of underspecification, where models with similar training performance diverge during deployment, resulting in unstable behaviors [54]. These limitations not only reduce the practical value of the models but also erode trust in their decisions, particularly in

high-reliability scenarios [55]. When a model fails, its black-box nature often makes it impossible to diagnose the root cause of the generalization error.

This context frames the core challenge of generalization as a multifaceted problem. At its center lies the tension between two competing objectives: learning features that are robust enough to be domain-invariant, while simultaneously being specific enough to remain discriminative for the task at hand. However, achieving a robust balance between these two is insufficient if the resulting model operates as an opaque 'black box,' as the lack of transparency prevents the diagnosis of failures and undermines trust in its decisions. Consequently, this study addresses three fundamental, interconnected problems: i) the identification and preservation of domain-invariant features, ii) the maintenance of discriminative features, and iii) the assurance of model interpretability. The relationship between these three core components is outlined in Figure 9.

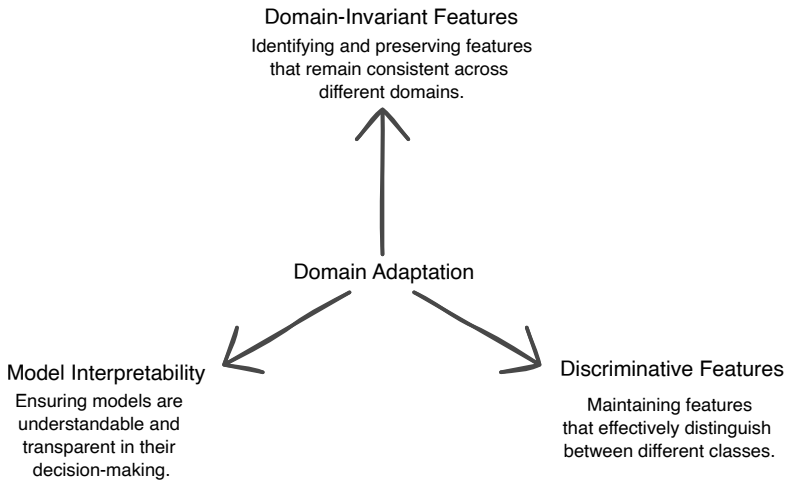


Figure 9: Problem overview. Graphical outline of the conceptual focus of this study.

4.1 Preservation of Domain-Invariant Features

Covariate shift poses a fundamental challenge, as it involves a change in the marginal distribution of features across domains, even when the conditional relationship remains unchanged [56]. This discrepancy impairs performance when applying the model to different settings, as source-domain representations may fail to capture valid structures in the target domain [57]. Even when an apparently invariant intermediate representation is learned, it may not retain task-relevant information, leading to systematic errors [58]. The problem is exacerbated by noise in features or labels from the source domain, or by samples affected by demographic imbalances or heterogeneous collection practices [59, 60]. In addition, insufficient representations that fail to preserve discriminative

information or shared support between domains compromise model robustness [61], as summarized in Figure 10.

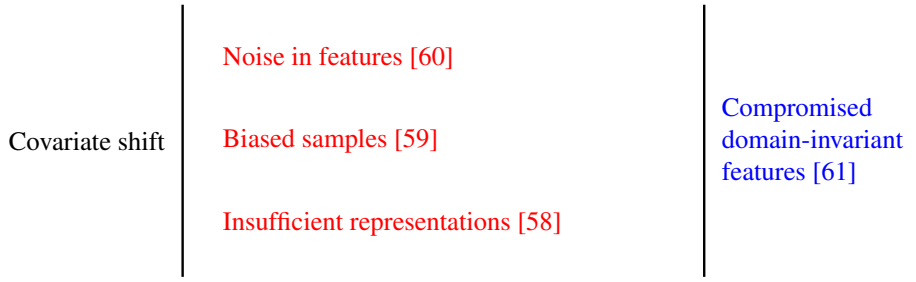


Figure 10: The diagram illustrates how the phenomenon of covariate shift manifests through specific contributing factors (causes, shown in red), which in turn lead to the primary consequence of compromised domain-invariant features (shown in blue).

Covariate shift also affects model stability by inducing systematic errors when test samples diverge from the training distribution [62]. This weakens generalization capacity, increases sensitivity to variation, and encourages overfitting [63]. The resulting lack of robustness limits applicability in unseen environments and hampers the detection of anomalous distributions, as models may assign high confidence to samples outside the original support [64, 65]. In EEG signals, covariate shift alters statistical properties without changing labels, thereby affecting prediction stability [66–68].

4.2 Maintenance of Discriminative Features

Generalization is also hindered when representations lack sufficient discriminative power to separate classes in the target domain [69]. Samples near cluster boundaries or far from class centers are more likely to be misclassified [70], and this loss of separability prevents the model from capturing the semantic structure of the domain, which reduces the reliability of predictions [71]. Maintaining discriminability is therefore essential to prevent errors caused by distributional mismatches [72].

A primary challenge to preserving discriminative power is concept shift, which occurs when the conditional probability distribution $P(y|\mathbf{x})$ changes between the source and target domains. In essence, the very definition of what makes a feature discriminative for a given class is altered, leading to a misalignment between the model's learned decision boundaries and the new semantic structure of the target data [73]. This phenomenon manifests in several ways that directly undermine feature separability. For instance, attempting marginal alignment of feature distributions can inadvertently merge distinct class clusters if the underlying label relationships have diverged [74]. Similarly, the use of poorly calibrated pseudo-labels can lead to distorted representations, as the model is trained on incorrect feature-label associations arising from the shift [75]. Furthermore, changes in data context can introduce noise-aware labels that are inconsistent with the source domain, further skewing the model's focus and reducing discriminability [76]. These mechanisms are summarized in Figure 11, which illustrates

how concept shift compromises the preservation of discriminative features through these contributing factors.

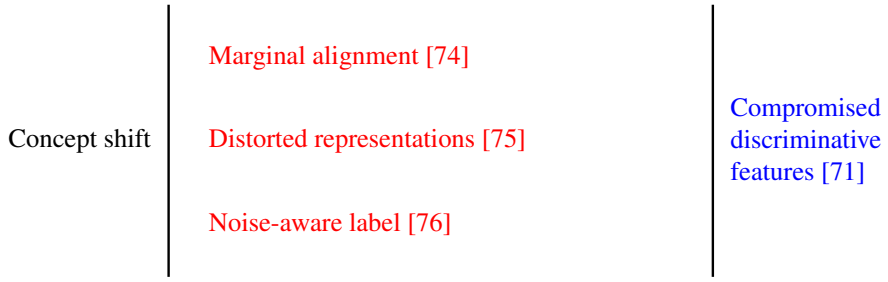


Figure 11: The diagram illustrates how the phenomenon of concept shift manifests through specific contributing factors (causes, shown in red), which in turn lead to the primary consequence of compromised discriminative features (shown in blue).

The loss of discriminability decreases model confidence and produces unstable decision boundaries, which increases classification errors [69]. In EEG applications, this leads to higher error rates for samples near boundaries and poor performance on unseen subjects [77]. It also increases semantic ambiguity and destabilizes predictions, making it harder to capture relevant patterns amid high inter-subject variability [78–80].

4.3 Interpretability

The challenge of generalization is critically intertwined with the issue of model interpretability. When a model fails in a new domain, as discussed previously, its 'black-box' nature makes it exceedingly difficult to diagnose the cause of the failure. This opacity creates a fundamental ambiguity: it becomes impossible to verify whether the learned representations correspond to meaningful semantic factors or merely to context-specific biases from the source domain [50]. Consequently, a lack of interpretability is not an independent issue but a direct barrier to building trustworthy systems, as it undermines the reliability of predictions in unknown domains [81]. This limitation is intensified by the technical challenges that cause this opacity, such as the absence of introspective mechanisms to analyze model behavior after adaptation [82]. When semantic differences are present, models tend to focus on irrelevant regularities [83]. In black-box models, interactions between latent subdomains cannot be traced [84]. Moreover, the lack of formal criteria for evaluating interpretability under domain shift contributes to uncertainty about the validity of the learned representations [85], as illustrated in Figure 12, which depicts the principal factors that compromise interpretability.

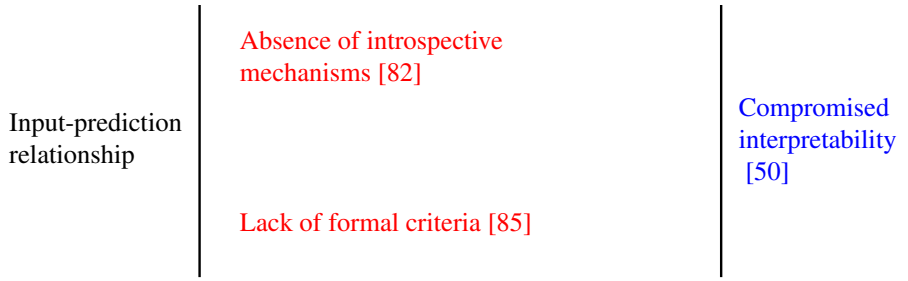


Figure 12: The diagram shows how the opaque input-prediction relationship is driven by specific contributing factors (causes, shown in red), leading to the direct consequence of compromised model interpretability (shown in blue).

The consequences of this opacity are especially critical in high-stakes application domains, where it is essential to verify whether model representations rely on genuinely relevant attributes or on spurious, domain-specific correlations—a challenge well-documented in fields like computer vision [86, 87]. This is particularly acute in the domain of EEG analysis, where the absence of interpretability hinders verification of whether decisions are based on genuine brain patterns or on signal artifacts [88]. This opacity prevents the detection of biases and systematic errors, a significant problem in contexts with high inter-subject variability [89, 90]. Furthermore, the lack of explanations that can integrate temporal, spatial, and spectral dimensions limits the validation of whether the model has learned coherent brain mechanisms, which is crucial for clinical and scientific trust [91].

4.4 Research Question

The preceding analysis has established that robust domain generalization is hindered by a tripartite challenge: the difficulty of learning domain-invariant features, the degradation of class-discriminative power, and the lack of model interpretability. This thesis confronts these interconnected issues within the specific context of domain adaptation for classification tasks. In this setting, the aforementioned challenges are often exacerbated by practical data conditions. For instance, the task of maintaining discriminative features is particularly intensified by problems such as overlapping labels and high inter-class similarity, while inadequate supervision can further obscure the learning of truly invariant and meaningful representations, making interpretability even more challenging.

This complex landscape thus converges into a central research question: **How can a domain adaptation framework be constructed to simultaneously improve generalization, class discrimination, and interpretability in deep learning models subjected to challenging classification scenarios with overlapping labels and inadequate supervision?**

5 State of the art

5.1 Classic Models

Classical domain adaptation methods are predominantly based on linear transformations of the feature space to mitigate the statistical discrepancy between source and target domains.

A seminal method is Correlation Alignment (CORAL), which aligns the covariance matrices of source and target domains via a linear transformation [92]. Given source and target sample matrices $\boldsymbol{X}_s \in \mathbb{R}^{M\times d}$ and $\boldsymbol{X}_t \in \mathbb{R}^{N\times d}$, with empirical covariances $\widehat{\Sigma}_s$ and $\widehat{\Sigma}_t$, respectively, the aligned features are computed as:

$$\boldsymbol{X}'_s = (\boldsymbol{X}_s - \hat{\mu}_s \mathbf{1})\,\widehat{\Sigma}_s^{-\frac{1}{2}}\,\widehat{\Sigma}_t^{\frac{1}{2}} + \hat{\mu}_t \mathbf{1}. \tag{31}$$

Here, M and N are the number of source and target samples, respectively, and d is the feature dimensionality. The term $\boldsymbol{X}'_s \in \mathbb{R}^{M\times d}$ is the transformed source data matrix; $\hat{\mu}_s, \hat{\mu}_t \in \mathbb{R}^{1\times d}$ are the row vectors representing the empirical means of the data; and $\mathbf{1}$ is a column vector of ones used for broadcasting.

Extensions such as CORAL++ and CORAL+ aim to improve robustness and adapt the parameter space [93–95]. However, the method is limited to second-order statistics, making it vulnerable to non-linear domain shifts [96].

To overcome such limitations, Subspace Alignment (SA) projects data into lower-dimensional subspaces and aligns their orthonormal bases [97]. Let $\boldsymbol{P}_s, \boldsymbol{P}_t \in \mathbb{R}^{d\times k}$ denote the matrices whose columns are the first $k \in \mathbb{N}$ (with $k < d$) principal components of the source and target domains, respectively. The alignment matrix is given by:

$$\boldsymbol{M} = \boldsymbol{P}_s^\top \boldsymbol{P}_t, \qquad \boldsymbol{Z}_s = \boldsymbol{X}_s \boldsymbol{P}_s \boldsymbol{M}, \qquad \boldsymbol{Z}_t = \boldsymbol{X}_t \boldsymbol{P}_t. \tag{32}$$

In this formulation, $\boldsymbol{M} \in \mathbb{R}^{k\times k}$ is the alignment matrix that maps the source subspace to the target subspace, and $\boldsymbol{Z}_s, \boldsymbol{Z}_t$ are the new data representations in the aligned subspaces.

Although SA has been extended to progressive and multi-source scenarios [98], its purely linear formulation limits its capacity to capture non-linear structures and class-specific boundaries [99, 100].

Enhancing the geometric fidelity of alignment, Geometric Transfer Learning (GTL) introduces a rotation matrix $\boldsymbol{R}$ that minimizes both geometric and statistical divergence:

$$\min_{\boldsymbol{R}} \|\boldsymbol{P}_s \boldsymbol{R} - \boldsymbol{P}_t\|_F^2 \quad \text{s.t.} \quad \boldsymbol{R}^\top \boldsymbol{R} = \boldsymbol{I}. \tag{33}$$

The optimization is performed over $\boldsymbol{R} \in \mathbb{R}^{k\times k}$, the optimal rotation matrix. The term $\|\cdot\|_F^2$ denotes the squared Frobenius norm, and $\boldsymbol{I}$ is the identity matrix of size $k \times k$.

Recent variants model complex manifolds via structured perturbations and geodesic metrics [101, 102]. Nevertheless, its reliance on linear subspaces continues to hinder adaptation under strong non-linear deformations.

To incorporate label information, Transfer Joint Matching (TJM) simultaneously reduces marginal and conditional distribution discrepancies via a unified projection matrix $\boldsymbol{A}$ [103]:

$$\min_{\boldsymbol{A}} \operatorname{Tr}(\boldsymbol{A}^\top \boldsymbol{X} \boldsymbol{M} \boldsymbol{X}^\top \boldsymbol{A}) + \lambda \|\boldsymbol{A}\|_F^2 \quad \text{s.t.} \quad \boldsymbol{A}^\top \boldsymbol{X} \boldsymbol{H} \boldsymbol{X}^\top \boldsymbol{A} = \boldsymbol{I}. \tag{34}$$

This objective function involves several components: $\boldsymbol{A} \in \mathbb{R}^{d \times k}$ is the adaptive projection matrix; $\boldsymbol{X} \in \mathbb{R}^{(M+N) \times d}$ is the matrix containing all samples; the TJM-specific matrix $\boldsymbol{M}$ encodes the joint distribution discrepancy; $\lambda > 0$ is a regularization hyperparameter; $\mathrm{Tr}(\cdot)$ is the trace operator; and $\boldsymbol{H}$ is a centering matrix.

TJM has been validated in multi-source and semi-supervised settings, including EEG classification [104, 105], but often produces ambiguous decision boundaries under class imbalance [77].

To address non-linearity, Transfer Component Analysis (TCA) maps input data into a Reproducing Kernel Hilbert Space (RKHS), where only marginal discrepancy is minimized:

$$\min_{\boldsymbol{A}} \ \mathrm{Tr}(\boldsymbol{A}^\top \boldsymbol{K} \boldsymbol{M}_0 \boldsymbol{K} \boldsymbol{A}) \quad \text{s.t.} \quad \boldsymbol{A}^\top \boldsymbol{K} \boldsymbol{H} \boldsymbol{K} \boldsymbol{A} = \boldsymbol{I}. \tag{35}$$

The key terms in this equation are $\boldsymbol{K} \in \mathbb{R}^{(M+N) \times (M+N)}$, representing the kernel matrix computed over all samples, and $\boldsymbol{M}_0$, the Maximum Mean Discrepancy (MMD) matrix that quantifies the marginal distribution shift.

TCA has proven useful in domains such as structural health monitoring and dynamic optimization [106, 107]. Nonetheless, its disregard for class-conditional structure limits its effectiveness in discriminative tasks [11].

Alternatively, Maximum Independence Domain Adaptation (MIDA) explicitly enforces domain invariance by maximizing statistical independence from the domain indicator variable:

$$\min_{\boldsymbol{A}} \ \mathrm{Tr}(\boldsymbol{A}^\top \boldsymbol{K} \boldsymbol{H} \boldsymbol{K} \boldsymbol{A}) - \mu \cdot \mathrm{Tr}(\boldsymbol{A}^\top \boldsymbol{K} \boldsymbol{D} \boldsymbol{K} \boldsymbol{A}). \tag{36}$$

This formulation utilizes a trade-off hyperparameter $\mu > 0$ and a diagonal matrix $\boldsymbol{D}$ containing domain labels, which is used to enforce independence between the learned features and the domain of origin.

This strategy has shown success in EEG decoding and pedestrian detection [108, 109], though its failure to align conditional distributions can produce invariant but semantically ambiguous features [110].

Finally, Joint Distribution Adaptation (JDA) extends the RKHS formulation to align both marginal and class-conditional distributions simultaneously:

$$\min_{\boldsymbol{A}} \ \mathrm{Tr}\left(\boldsymbol{A}^\top \boldsymbol{K} \left(\boldsymbol{M}_0 + \sum_{c=1}^{C} \boldsymbol{M}_c \right) \boldsymbol{K} \boldsymbol{A} \right) \quad \text{s.t.} \quad \boldsymbol{A}^\top \boldsymbol{K} \boldsymbol{H} \boldsymbol{K} \boldsymbol{A} = \boldsymbol{I}. \tag{37}$$

The summation is performed over the total number of classes C, with $\boldsymbol{M}_c$ being the MMD matrix for the conditional distribution of class c, thereby allowing for class-aware alignment.

JDA has been applied to dynamic system modeling and fuzzy regression [111, 112]. However, its reliance on static weighting of marginal and conditional terms can impair its performance under class imbalance or within-class variability [113, 114]. A comparative summary of these classical approaches, categorized by their underlying assumptions and transformation strategies, is provided in Figure 13.

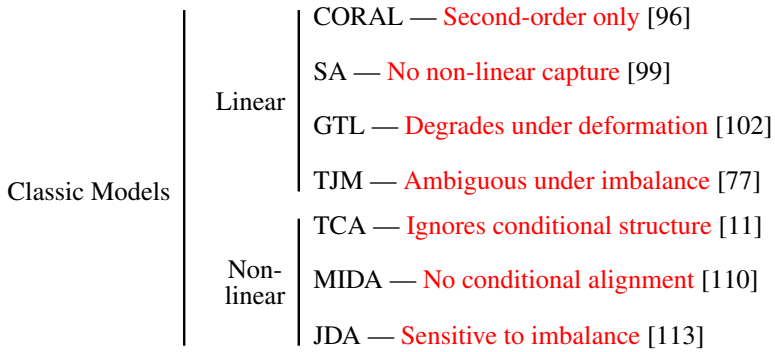


Figure 13: Fundamental limitations of Classic Models for domain adaptation

5.2 Preservation of Domain-Invariant Features

In contrast to classical models, deep learning approaches integrate feature extraction and domain adaptation into a unified, end-to-end optimization process. These methods typically involve three core components: a feature extractor, $\mathscr{F} : \mathscr{X} \to \mathbb{R}^d$; a task-specific classifier, $\mathscr{G} : \mathbb{R}^d \to [0,1]^C$; and often a domain discriminator, $\mathscr{D} : \mathbb{R}^d \to [0,1]$. The goal is to leverage the representational capacity of these networks to construct a shared latent space where the discrepancy between domains is actively minimized.

Adversarial methods, exemplified by the Domain-Adversarial Neural Network (DANN), aim to induce domain invariance through a minimax game. The feature extractor $\mathscr{F}$ is trained to deceive the domain discriminator $\mathscr{D}$, which simultaneously learns to distinguish the origin of the samples. The joint optimization over the network parameters $\theta_{\mathscr{F}}$, $\theta_{\mathscr{G}}$, and $\theta_{\mathscr{D}}$ is expressed as:

$$\min_{\theta_{\mathscr{F}},\theta_{\mathscr{G}}} \max_{\theta_{\mathscr{D}}} \mathscr{L}_{\text{sup}}\big(\mathscr{G}(\mathscr{F}(\mathscr{D}^s)),\mathbf{y}^s\big) + \lambda\, \mathscr{L}_{\text{ad}}\big(\mathscr{D}(\mathscr{F}(\mathscr{D}^s \cup \mathscr{D}^t))\big),$$

where $\mathscr{L}_{\text{sup}}$ is the supervised classification loss on the labeled source domain $\mathscr{D}^s = \{\mathbf{x}_i^s, \mathbf{y}_i^s\}_{i=1}^{N_s}$, and $\mathscr{L}_{\text{ad}}$ is the adversarial loss for domain classification over both source and unlabeled target data $\mathscr{D}^t = \{\mathbf{x}_j^t\}_{j=1}^{N_t}$. The hyperparameter $\lambda > 0$ balances the two objectives. The overall goal is to learn representations that are simultaneously discriminative and domain-indistinguishable [115]. This paradigm has shown effectiveness across diverse application areas, including natural language processing, medical image segmentation, and industrial scenarios [116–120]. Nevertheless, it can lead to negative transfer due to incorrect semantic pairings, particularly when the multimodal structure of the data is complex [121, 122].

To mitigate these issues, architectures such as Adversarial Discriminative Domain Adaptation (ADDA) have been proposed, which decouple the feature extractors ($\mathscr{F}^s, \mathscr{F}^t$) and adopt asymmetric training [123]. Several variants extend ADDA to semi-supervised (SADDA), few-shot (T-ADDA), and noisy scenarios (U-ADDA) [124–126]. Another direction, represented by ALDA, integrates self-labeling and error correction

during adaptation [127]. Despite these advances, such methods often yield ambiguous representations, ultimately reducing the discriminative capacity of the model [128, 129].

To enforce semantic consistency, Discriminative Adversarial Domain Adaptation (DADA) incorporates both class prediction and domain discrimination within a unified architecture [130]. This approach has been effectively applied to complex systems in energy and industrial domains [131, 132]. However, its optimization may overemphasize domain invariance at the expense of inter-class separability [133].

Parallel to adversarial strategies, an alternative family of methods combines supervised learning with explicit statistical alignment. Defining the feature extractor as a composition of layers $\mathscr{F} = (\breve{f}_L \circ \cdots \circ \breve{f}_1)$, the general formulation for these methods is:

$$\min_{\theta_{\mathscr{F}}, \theta_{\mathscr{G}}} \mathscr{L}_{\text{sup}}\big(\mathscr{G}(\mathscr{F}(\mathscr{D}^s)), \mathbf{y}^s\big) + \alpha \sum_{l \in \mathscr{L}} \mathscr{L}_{\text{stat}}\big(\mathscr{F}_l(\mathscr{D}^s), \mathscr{F}_l(\mathscr{D}^t)\big),$$

where $\mathscr{L}_{\text{stat}}$ is a statistical loss function quantifying the discrepancy between domain activations. The term $\mathscr{F}_l(\cdot)$ denotes the feature representation at an intermediate layer l, the summation is performed over a set of selected layers $\mathscr{L}$, and $\alpha > 0$ is a weighting hyperparameter.

Within this framework, Deep CORAL extends covariance alignment to deep features, embedding discrepancy reduction directly into the training process [134]. It has been applied in fault diagnosis, medical imaging, and scenarios involving differential privacy [10, 135, 136]. A known limitation of this approach is the generation of conflicting gradients, which can hinder both convergence stability and alignment effectiveness [137].

Similarly, the Maximum Mean Discrepancy (MMD) has been widely adopted as the $\mathscr{L}_{\text{stat}}$ term to align feature distributions in a Reproducing Kernel Hilbert Space (RKHS) [138, 139]. Conditional extensions such as LMMD enable per-class alignment of subdomains [140]. A persistent issue with these methods is semantic collapse, where distinct classes may be improperly mapped to overlapping regions, impairing the model's discriminative capabilities in the target domain [141].

Finally, the Deep Joint Adaptation Network (DJAN) constitutes one of the most comprehensive frameworks, seeking to jointly align marginal and conditional distributions, often in conjunction with adversarial objectives [142–144]. Nonetheless, its reliance on pseudo-labels ($\hat{\mathbf{y}}^t$) to estimate target conditional distributions introduces potential noise, particularly in settings with large domain shifts, leading to spurious semantic alignments that can degrade performance [145].

The architectural distinctions and limitations of domain-invariant models are visually outlined in Figure 14.

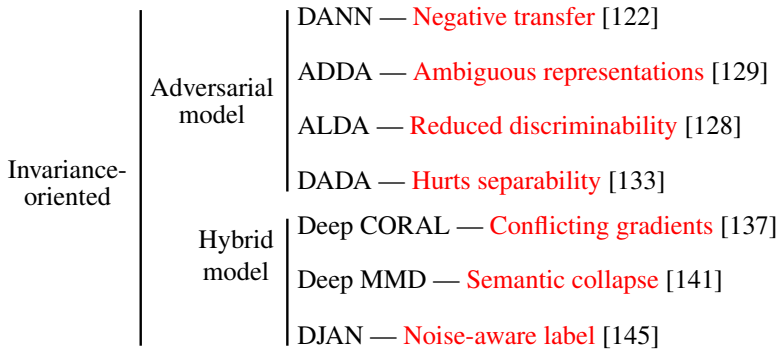


Figure 14: Diagram illustrating the key drawbacks, such as catastrophic forgetting and sensitivity to data artifacts, in techniques aimed at Preserving Domain-Invariant Features

5.3 Maintenance of Discriminative Features

While the adversarial and statistical alignment methods discussed in the previous section are effective at learning domain-invariant representations, an overemphasis on this objective can lead to a critical drawback: the degradation of discriminative class structure. This deficiency arises when, in the process of removing domain-specific information, the model also discards features essential for inter-class separability. The result is latent representations that, although globally aligned, may conflate semantically distinct classes, thereby compromising predictive performance. To counteract this, a complementary family of strategies aims to explicitly preserve or induce discriminative structure during adaptation.

These approaches can be generally formalized as a joint optimization problem over the parameters of source-specific and target-specific models. Let $\mathscr{F}^s$ and $\mathscr{F}^t$ be the feature extractors for the source and target domains, and $\mathscr{G}^s$ and $\mathscr{G}^t$ be their respective classifiers. The objective is to learn the model parameters $(\theta_{\mathscr{F}^s}, \theta_{\mathscr{F}^t}, \theta_{\mathscr{G}^s}, \theta_{\mathscr{G}^t})$ by minimizing:

$$\min_{\theta_{\mathscr{F}^s}, \theta_{\mathscr{F}^t}, \theta_{\mathscr{G}^s}, \theta_{\mathscr{G}^t}} \mathscr{L}_{\text{sup}}\big(\mathscr{G}^s(\mathscr{F}^s(\mathscr{D}^s)), \mathbf{y}^s\big) + \beta\, \mathscr{L}_{\text{transfer}}\big(\mathscr{D}^s, \mathscr{D}^t; \theta_{\mathscr{F}^s}, \theta_{\mathscr{F}^t}, \theta_{\mathscr{G}^s}, \theta_{\mathscr{G}^t}\big),$$

where $\mathscr{L}_{\text{transfer}}$ is a transfer loss that encourages discriminative alignment, and $\beta > 0$ is a trade-off hyperparameter. This loss term encompasses mechanisms such as fine-tuning, collaborative learning, or conditional alignment.

A primary line of work focuses on direct transfer strategies, notably fine-tuning, which adapts a pre-trained source model $(\mathscr{F}^s, \mathscr{G}^s)$ using annotations from the target domain [146–148]. Its main limitation is the degradation of learned representations (catastrophic forgetting), which undermines generalization [149]. In unsupervised settings, Teacher–Student frameworks use the source model as a "teacher" to generate pseudo-labels $\hat{\mathbf{y}}^t = \mathscr{G}^s(\mathscr{F}^s(\mathbf{x}^t))$ that guide the adaptation of a "student" model $(\mathscr{F}^t, \mathscr{G}^t)$ on the target domain [150–152]. However, the performance of such methods depends heavily on pseudo-label quality, rendering them susceptible to noise, particularly with

high inter-domain discrepancy [153]. More recently, foundation models have opened new avenues for adaptation via mechanisms such as prompt tuning or adapter modules [154, 155], though their deployment remains constrained by semantic misalignment and high computational cost [156].

To jointly optimize alignment and classification in a more principled way, meta-learning-based approaches have emerged, aiming to balance these objectives through coherence maximization [157]. For example, ToAlign decomposes features into relevant and irrelevant components for selective alignment, while other methods adopt test-time adaptation or multi-objective formulations [158–160]. Despite their sophistication, these techniques risk overfitting to the alignment meta-task, which can hinder generalization [161].

Another research line integrates knowledge distillation to facilitate efficient cross-domain transfer [162]. Some approaches leverage reinforcement learning to identify reliable samples, while others adopt direct distillation via adapters that separate domain-invariant and domain-specific components [163]. Self-distillation strategies have also been proposed to enhance discriminability without auxiliary models [164]. Nonetheless, these methods remain vulnerable to alignment errors, which may propagate biases and impair generalization [165].

Finally, conditional alignment methods offer a direct mechanism for preserving semantic structure by incorporating class information into the domain alignment process. The Conditional Domain Adversarial Network (CDAN) exemplifies this approach by conditioning domain discrimination on the joint distribution of features and predicted labels. Assuming shared feature extractor $\mathscr{F}$ and classifier $\mathscr{G}$, its objective is:

$$\min_{\theta_{\mathscr{F}},\theta_{\mathscr{G}}} \max_{\theta_{\mathscr{D}}} \mathscr{L}_{\text{sup}}\big(\mathscr{G}(\mathscr{F}(\mathscr{D}^s)),\mathbf{y}^s\big) \;+\; \gamma\,\mathscr{L}_{\text{c-ad}}\left(\mathscr{D},\mathscr{F},\mathscr{G};\mathscr{D}^s,\mathscr{D}^t\right),$$

where $\mathscr{L}_{\text{c-ad}}$ is the conditional adversarial loss. This loss is applied to the output of a discriminator $\mathscr{D}$ that receives the tensor product $\mathscr{F}(\mathbf{x}) \otimes \mathscr{G}(\mathscr{F}(\mathbf{x}))$ as input, with $\otimes$ denoting the tensor product operator and $\gamma > 0$ as a weighting hyperparameter. Extensions of CDAN have introduced confidence-aware weighting and adjustments for class imbalance [166–169]. While CDAN effectively models multimodal structures, it is sensitive to low-confidence predictions, which can limit its robustness [133]. In parallel, self-supervised learning (SSL) has been introduced into domain adaptation pipelines, typically by leveraging contrastive objectives to learn transferable features without explicit supervision [170–172]. Yet, the efficacy of SSL can be undermined by semantic discrepancies and high computational burden [156]. In response, information-theoretic approaches optimize transfer through objectives based on mutual information or entropy [173, 174]. By leveraging strategies such as entropy minimization, these methods regularize latent representations and improve generalization [175–177]. Nevertheless, their applicability is often hampered by the challenges of robustly estimating information-theoretic quantities [178]. Figure 15 summarizes the principal methods aimed at preserving class discriminability, along with their known challenges.

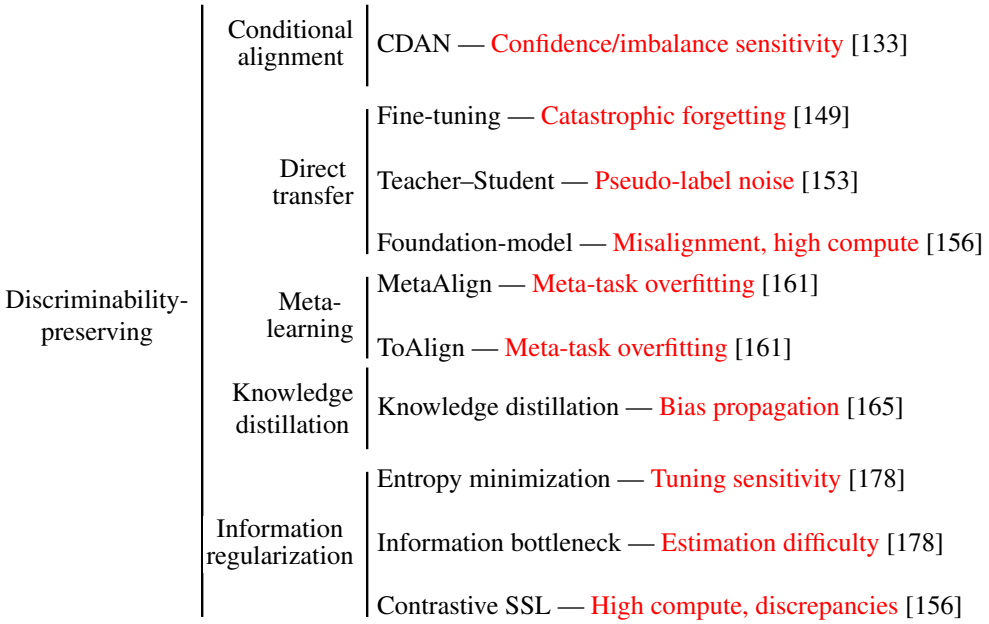


Figure 15: Key challenges in Maintaining Discriminative Features across different domain adaptation algorithms.

5.4 Interpretability

Beyond generalization and discriminability, interpretability has emerged as a fundamental pillar in domain adaptation, particularly in high-risk applications where understanding the model's behavior is essential for ensuring trust and transparency [179].

A first avenue toward interpretability involves the analysis of the latent space, with the goal of visualizing the structure of learned representations to evaluate the effectiveness of the alignment. Linear techniques such as Principal Component Analysis (PCA), although computationally efficient, are limited in their ability to capture the non-linear relationships that often underlie domain discrepancies [180]. In contrast, non-linear methods such as t-distributed Stochastic Neighbor Embedding (t-SNE) and Uniform Manifold Approximation and Projection (UMAP) have proven more effective in representing complex geometries [181]. UMAP, in particular, stands out for its capacity to preserve both local and global structure, its robustness to parameter variations, and its scalability, making it a suitable tool for visualizing semantic alignment across domains [182, 183].

A second interpretability strategy focuses on post-hoc attribution methods, which aim to explain individual predictions without modifying the model architecture. Gradient-based techniques such as Grad-CAM are widely used due to their simplicity in generating visual attention maps. However, their effectiveness decreases in deeper architectures,

prompting the development of Grad-CAM++, which improves spatial precision by incorporating higher-order derivatives [158, 184]. Despite these advancements, Grad-CAM++ remains limited when applied to models with non-linear activation functions [185]. In the context of domain adaptation, it has been used to visually validate semantic consistency by assessing whether the activated regions in the target domain correspond to those in the source [186].

Additional methods, such as Layer-wise Relevance Propagation (LRP) and Shapley Additive Explanations (SHAP), offer more rigorous quantification by assigning relevance scores to input features. These scores facilitate the detection of spurious patterns or conflicting decision logic [187, 188]. LRP redistributes the model's output backward across layers, while SHAP, grounded in cooperative game theory, decomposes the prediction into the marginal contribution of each feature. Despite the availability of such tools, a fundamental gap persists in the literature: the lack of interpretability methods explicitly designed for domain adaptation. This absence underscores the need for explanatory frameworks that not only validate the outcomes of transfer learning but also elucidate the mechanisms underlying adaptation itself—a critical requirement for the safe and transparent deployment of these models in sensitive domains [189]. An overview of interpretability techniques and their limitations within the context of domain adaptation is presented in Figure 16.

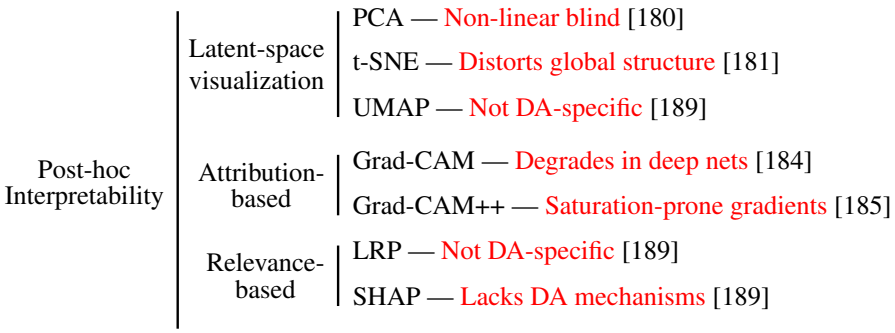


Figure 16: Challenges and limitations of prominent Interpretability methods in machine learning.

5.5 Summary

The literature review reveals a significant evolution in domain adaptation methodologies, progressing from classical linear transformations to sophisticated deep learning architectures. Classical approaches, such as Correlation Alignment (CORAL) and Subspace Alignment (SA), established foundational principles for aligning statistical moments across domains. More advanced methods like Joint Distribution Adaptation (JDA) [111] offer more robust solutions by simultaneously aligning both marginal and class-conditional distributions. However, the efficacy of these classical techniques

remains constrained by underlying assumptions of linearity, limiting their capacity to handle complex, non-linear domain shifts and inherent semantic ambiguities.

In response, deep learning-based methods have emerged, offering end-to-end frameworks that learn both feature extraction and domain alignment simultaneously. One category of these methods focuses on preserving domain-invariant features, using adversarial training (DANN) [115] or statistical alignment (MMD) [138] to reduce domain discrepancy. Nonetheless, these approaches are susceptible to critical failure modes, such as negative transfer, where alignment degrades performance, and semantic collapse, where distinct classes are improperly mapped to overlapping regions. To address these shortcomings, a second category of methods aims to explicitly preserve discriminative power. Among these, the Conditional Domain Adversarial Network (CDAN) [166, 167] stands out as a robust approach, conditioning the alignment on class predictions to maintain semantic structure. However, even advanced strategies like CDAN, teacher-student models, and meta-learning often struggle with noisy pseudo-labels, catastrophic forgetting, and instabilities arising from conflicting optimization objectives, which compromise the semantic integrity of the learned representations.

Furthermore, a significant gap persists in the interpretability of these adaptation mechanisms. This limitation is particularly acute in high-stakes applications such as EEG signal analysis, where transparent validation requires explanations across the spatial, temporal, and spectral dimensions of the data. While post-hoc methods like Grad-CAM [158] offer valuable spatial insights, they often fail to capture the dynamic temporal and rich spectral patterns crucial for neurophysiological interpretation. This absence of multi-faceted interpretability makes it difficult to diagnose adaptation failures or verify that alignment is based on meaningful neurophysiological correspondence rather than spurious correlations. This thesis is therefore motivated by the development of a framework that not only achieves robust, class-aware domain alignment but also provides interpretable, multi-dimensional validation of its underlying adaptation logic.

6 Objectives

6.1 General objective

Develop deep learning base domain adaptation framework to address the challenges of generalization, class discrimination, and interpretability in for classification task models under conditions of overlapping labels, structural noise, and limited supervision, with an emphasis on the stability and traceability of behavior in unseen domains.

6.2 Specific objectives

1 Develop a regularization mechanism to enhance model robustness against target domain uncertainty, considering the effect of overlapping labels and noisy pseudo-labels in limited supervision settings.

2 Design an alignment strategy that preserves semantic consistency across domains, in order to reduce the class-level distributional discrepancy and retain discriminative information in the presence of covariate shift.

3 Establish an interpretability approach to facilitate the analysis of the semantic coherence of representations, through the qualitative evaluation of activated regions and their correspondence with relevant attributes, in scenarios characterized by high inter-class ambiguity and inter-domain variability.

7 Outline and Contributions

In this section, we provide a concise overview of the main contributions presented in this thesis proposal. Our work focuses on a novel framework, termed Conditional α-Rényi's Entropy-based Domain Adaptation (CREDA), designed as a comprehensive solution to address the fundamental challenges of generalization, class discrimination, and interpretability in unsupervised domain adaptation scenarios. These scenarios are often characterized by overlapping labels, structural noise, and limited supervision. Figure 17 illustrates the main pillars of our proposed solution.

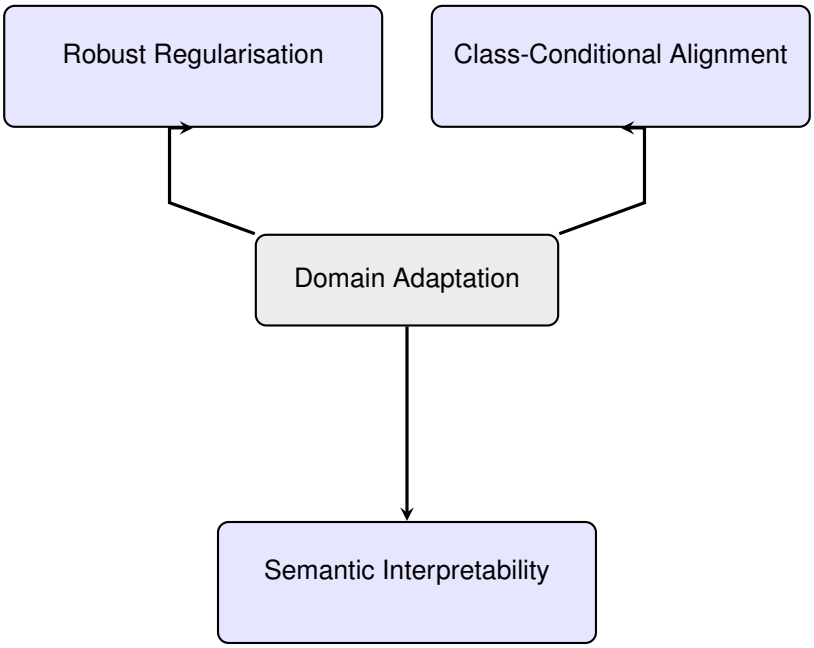


Figure 17: Schematic diagram illustrating the proposed solutions for generalization, discrimination, and interpretability. The framework addresses target domain uncertainty through an entropy-based weighting mechanism, preserves semantic structure via class-conditional alignment, and ensures traceability through qualitative analysis of learned representations.

7.1 A Robust Regularization Mechanism for Mitigating Target Domain Uncertainty

To address the challenge of performance degradation due to noisy or ambiguous pseudo-labels in the target domain, we introduce a novel regularization mechanism. This contribution is founded on a kernel-based quadratic Rényi entropy estimator, which allows for the non-parametric quantification of predictive uncertainty. We develop a confidence weighting scheme that dynamically modulates the influence of each target sample on the alignment objective. This approach enhances model robustness by systematically attenuating the impact of low-confidence predictions, which is particularly critical in limited supervision settings where pseudo-label quality can be unreliable.

7.2 A Class-Conditional Alignment Strategy for Preserving Semantic Coherence

To overcome the limitations of marginal distribution alignment, which often leads to semantic collapse and impairs discriminative power, we propose a strategy for class-conditional alignment. This method leverages a kernel-based mutual information estimator, derived from the joint and marginal Rényi's entropies, to explicitly minimize the discrepancy between class-conditional distributions of the source and target domains. By integrating this estimator as a regularization term, our CREDA framework enforces semantic consistency, ensuring that the learned feature representations are not only domain-invariant but also preserve the discriminative structure necessary to distinguish between classes, even in the presence of significant covariate shift.

7.3 An Interpretability Framework for Analyzing Semantic Coherence

Recognizing the critical need for model transparency, we establish an interpretability approach to validate the behavior of the adaptation framework. This contribution provides a methodology for the qualitative and quantitative analysis of the learned representations. We employ two key techniques:

1. Latent space visualization using non-linear dimensionality reduction (e.g., UMAP) to inspect the semantic clustering and inter-domain mixing of learned features.

2. Post-hoc attribution analysis (e.g., Grad-CAM++) to examine the model's decision-making process by identifying the input regions that are most salient for a given class prediction across domains.

This dual approach facilitates the verification of semantic coherence, ensuring that domain alignment is achieved without compromising the model's reliance on relevant, class-specific attributes.

8 Methodology

8.1 Kernel Methods Fundamentals

Kernel methods provide a powerful framework for developing nonlinear algorithms. The core idea is to implicitly map the input data from its original space $\mathscr{X}$ into a high-dimensional, or even infinite-dimensional, feature space $\mathscr{H}$ via a nonlinear mapping $\Phi : \mathscr{X} \to \mathscr{H}$. The space $\mathscr{H}$ is a special type of Hilbert space known as a Reproducing Kernel Hilbert Space (RKHS), and the mapping Φ is chosen such that complex patterns in the data may become simpler, e.g., linearly separable $\mathscr{H}$ [190].

Explicitly computing the coordinates of the mapped data points $\Phi(x)$ is often computationally expensive or infeasible. Then, the kernel trick allows us to bypass this by defining a kernel function $\kappa : \mathscr{X} \times \mathscr{X} \to \mathbb{R}$ that computes the inner product between two points in the feature space:

$$\kappa(x_i, x_j) = \langle \Phi(x_i), \Phi(x_j) \rangle_{\mathscr{H}}. \tag{38}$$

Then, we work directly with the kernel function without ever needing to know the explicit form of Φ or the structure of $\mathscr{H}$. Indeed, an RKHS is uniquely defined by this property, ensuring that all computations can be performed using the kernel [41]. In practice, a common choice for the kernel function is the Gaussian kernel:

$$\kappa_\sigma(x_i, x_j) = \exp\left(\frac{-\|x_i - x_j\|_2^2}{2\sigma^2}\right), \tag{39}$$

which corresponds to an infinite-dimensional feature space, with $\sigma \in \mathbb{R}^+$. Still, its mathematical tractability and intuitive notion of similarity make it a commonly used approach [191].

8.2 Kernel-Based α-Rényi's Entropy Estimation

Let X be a continuous random variable with a probability density function (PDF) $f(x)$, $x \in X$, the Rényi's α-order entropy is defined as follows [192]:

$$H_\alpha(X) = \frac{1}{1-\alpha} \log \int_{\mathscr{X}} f(x)^\alpha dx, \tag{40}$$

where $\alpha > 0$, and $\alpha \neq 1$. A primary challenge in applying this definition is that in most practical scenarios, especially with high-dimensional data like deep features, the underlying PDF $f(x)$ is unknown [193]. To circumvent this, a Parzen-window method, also known as Kernel Density Estimation (KDE) can be employed. Namely, given a finite set of N samples $\{x_i \in X\}_{i=1}^N$, the PDF at any point x can be estimated as the average of kernel functions centered at each sample [194]:

$$\hat{f}(x) = \frac{1}{N} \sum_{i=1}^{N} \kappa_\sigma(x, x_i), \tag{41}$$

where the Gaussian kernel is selected for its mathematical simplicity and desirable smoothing behavior (see Equation (39)). In particular, when $\alpha = 2$ in Equation (40), we

focus on the special case of Rényi's entropy, known as quadratic entropy. Indeed, the integral term in Equation (40), $\int f(x)^2 dx$, is known as the Information Potential (IP) [195], a measure of the average information contained in the distribution. Substituting the KDE estimator $\hat{f}(x)$ into the IP integral, yields the following:

$$\begin{aligned} \hat{V}_2(X) &= \int \hat{f}(x)^2 \, dx = \int \left(\frac{1}{N} \sum_{i=1}^{N} \kappa_\sigma(x, x_i) \right) \left(\frac{1}{N} \sum_{j=1}^{N} \kappa_\sigma(x, x_j) \right) dx \\ \hat{V}_2(X) &= \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \int \kappa_\sigma(x, x_i) \, \kappa_\sigma(x, x_j) \, dx \end{aligned} \tag{42}$$

A significant advantage of using a Gaussian kernel is that the integral in Equation (42) has a closed-form solution based on the convolution property of Gaussians [196]:

$$\int \kappa_\sigma(x, x_i) \kappa_\sigma(x, x_j) dx = \kappa_{\sqrt{2}\sigma}(x_i, x_j). \tag{43}$$

The latter simplifies the IP estimator to a practical, sample-based formula that depends only on pairwise interactions between samples, completely bypassing the need for explicit PDF estimation:

$$\hat{V}_2(X) = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \kappa_{\sqrt{2}\sigma}(x_i, x_j). \tag{44}$$

Next, let $\mathbf{K} \in \mathbb{R}^{N \times N}$ be a Gram matrix whose elements are the pairwise kernel evaluations, $\mathbf{K}_{ij} = \kappa_{\sqrt{2}\sigma}(x_i, x_j)$. The sum of all elements in this matrix can be computed as $\mathbf{1}^T \mathbf{K} \mathbf{1}$, where $\mathbf{1}$ is a column vector of ones. This gives a matrix-based estimator for the IP:

$$\hat{V}_2(X) = \frac{1}{N^2} \mathbf{1}^\top \mathbf{K} \mathbf{1} \tag{45}$$

Recently, a α-Rényi matrix-based operator extracts from the IP expression in Equation (45). More generally, Rényi's entropy can be defined directly over the eigenspectrum of a normalized Gram matrix. If we define a normalized Gram matrix $\mathbf{A} = \mathbf{K}/\mathrm{tr}(\mathbf{K})$, where $\mathrm{tr}(\cdot)$ is the trace operator, the entropy is given by [197]:

$$H_\alpha(\mathbf{A}) = \frac{1}{1-\alpha} \log(\mathrm{tr}(\mathbf{A}^\alpha)) = \frac{1}{1-\alpha} \log \left(\sum_i \breve{\lambda}_i(\mathbf{A})^\alpha \right), \tag{46}$$

where $\breve{\lambda}_i(\mathbf{A})$ are the eigenvalues of $\mathbf{A}$. For our work with $\alpha = 2$, we use a computationally stable form based on the Frobenius norm: $H_2(\mathbf{A}) = -\log(\mathrm{tr}(\breve{\mathbf{A}}^\top \breve{\mathbf{A}}))$, where $\breve{\mathbf{A}} = \mathbf{A}/\mathrm{tr}(\mathbf{A})$, $\mathrm{tr}(\breve{\mathbf{A}}) = 1$, and $\|\mathbf{A}\|_F^2 = \mathrm{tr}(\mathbf{A}^\top \mathbf{A})$. This matrix-based formulation is essential for deep learning due to several key properties:

– Non-parametric: It makes no prior assumptions about the underlying data distribution, making it highly suitable for the complex and high-dimensional feature spaces learned by neural networks.

– Differentiable: The entropy loss is a function of the Gram matrix elements, which are themselves differentiable functions of the feature vectors produced by a given network. This allows gradients to be backpropagated through the kernel computations to the network's parameters, enabling end-to-end training.

– Robust: The entropy is calculated based on the collective geometric structure of the data, as captured by all pairwise interactions in the Gram matrix. This makes the measure inherently robust to outliers, which would have a limited impact on the overall sum of kernel values.

The matrix-based entropy framework in Equation (46) can be extended to measure relationships between two random variables, X and Y, represented by paired feature vectors $\{\mathbf{f}_{X,i}, \mathbf{f}_{Y,i}\}_{i=1}^{N}$. This is achieved by defining a joint Gram matrix using the Hadamard (element-wise) product as follows:

– Joint Entropy—(JE). Let $\mathbf{K}_X \in \mathbb{R}^{N\times N}$ and $\mathbf{K}_Y \in \mathbb{R}^{N\times N}$ be the Gram matrices computed from the feature sets of X and Y, respectively. The joint entropy based on the α-Rényi estimator is defined as follows [198]:

$$H_\alpha(\mathbf{K}_X, \mathbf{K}_Y) = \frac{1}{1-\alpha} \log\left(\mathrm{tr}(\breve{\mathbf{K}}_{X,Y})^\alpha\right), \tag{47}$$

where $\mathbf{K}_{XY} = \mathbf{K}_X \odot \mathbf{K}_Y$, $\breve{\mathbf{K}}_{X,Y} = \mathbf{K}_{X,Y}/\mathrm{tr}(\mathbf{K}_{X,Y})$, and $\odot$ denotes the Hadamard product. Of note, the joint matrix $\mathbf{K}_{XY}$ captures the similarity between pairs of samples in the joint feature space.

– Mutual Information—(MI). It quantifies the statistical dependence between two variables. In the matrix-based framework, it is defined in analogy to its classic information-theoretic definition:

$$I_\alpha(\mathbf{K}_X; \mathbf{K}_Y) = H_\alpha(\mathbf{K}_X) + H_\alpha(\mathbf{K}_Y) - H_\alpha(\mathbf{K}_X, \mathbf{K}_Y), \tag{48}$$

where each entropy term is computed from its respective (normalized) Gram matrix. Maximizing MI is a common objective in representation learning, as it encourages a representation to retain information about a relevant variable.

– Conditional Entropy—(CE). It measures the remaining uncertainty in a variable X given that Y is known. It is defined as follows:

$$H_\alpha(\mathbf{K}_X|\mathbf{K}_Y) = H_\alpha(\mathbf{K}_X, \mathbf{K}_Y) - H_\alpha(\mathbf{K}_Y) \tag{49}$$

Minimizing conditional entropy is equivalent to making X more predictable from Y.

8.3 Domain Adaptation with α-Rényi Entropy-Based Label Weighting and Regularization

Our proposed method, Conditional α-Rényi's Entropy Regularization (CREDA), is designed for end-to-end training in unsupervised domain adaptation. The framework

leverages a deep feature extractor $\mathscr{F} : \mathscr{X} \to \mathbb{R}^d$ that maps an input image $\mathbf{x} \in \mathbb{R}^{\check{H}\times\check{W}\times\check{C}}$, with $\mathscr{X} \subseteq \mathbb{R}^{p'}$, $p' = \check{H}\times\check{W}\times\check{C}$, to a d-dimensional feature vector $\mathbf{f} \in \mathbb{R}^d$, as follows:

$$\mathbf{f} = \mathscr{F}(\mathbf{x}) = (\check{f}_L \circ \check{f}_{L-1} \circ \cdots \circ \check{f}_1)(\mathbf{x}), \tag{50}$$

where $\check{f}_l(\cdot)$ stands for the l-th feature extractor layer ($l \in \{1,\ldots,L\}$), and $\circ$ is the function composition operator. Moreover, a classifier $\mathscr{G} : \mathbb{R}^d \to [0,1]^C$ that predicts class-probability vector $\mathbf{g} \in [0,1]^C$, is defined as follows:

$$\mathbf{g} = \mathscr{G}(\mathbf{f}) = (\check{g}_L \circ \check{g}_{L-1} \circ \cdots \circ \check{g}_1)(\mathbf{f}), \tag{51}$$

with $\check{g}_{l'}(\cdot)$ as a given classifier layer ($l' \in \check{L}$), $\sum_{c=1}^{C} g_c = 1$, and $g_c \in \mathbf{g}$.

In practice, we are given a labeled source domain $\mathscr{D}^s = \{\mathbf{x}_i^s \in \mathbb{R}^{p'}, \mathbf{y}_i^s \in \{0,1\}^C\}_{i=1}^{N_s}$, with $\sum_{c=1}^{C} y_{i,c}^s = 1$, $y_{i,c}^s \neq y_{i,c'}^s$, $c, c' \in C$, and $y_{i,c}^s, y_{i,c'}^s \in \mathbf{y}_i^s$. Also, an unlabeled target domain is provided as $\mathscr{D}^t = \{\mathbf{x}_j^t \in \mathbb{R}^{p'}\}_{j=1}^{N_t}$. For each class c, we compute the source, target, and source-target kernel-based matrices $\mathbf{K}_c^s \in \mathbb{R}^{n_c^s \times n_c^s}$, $\mathbf{K}_c^t \in \mathbb{R}^{n_c^t \times n_c^t}$, and $\mathbf{K}_c^{st} \in \mathbb{R}^{n_c^t \times n_c^s}$, as follows:

$$\mathbf{K}_c^s = [\kappa_{\sigma_s}(\mathbf{f}_i^s, \mathbf{f}_{i'}^s)], \quad \forall i, i' \in n_c^s : \quad \mathbf{f}_i^s = \mathscr{F}(\mathbf{x}_i^s), \quad y_{i,c}^s = 1 \tag{52}$$

$$\mathbf{K}_c^t = [\kappa_{\sigma_t}(\mathbf{f}_j^t, \mathbf{f}_{j'}^t)], \quad \forall j, j' \in n_c^t : \quad \mathbf{f}_j^t = \mathscr{F}(\mathbf{x}_j^t), \quad \arg\max_{c'} g_{j,c'}^t = c \tag{53}$$

$$\mathbf{K}_c^{st} = [\kappa_{\sigma_{st}}(\mathbf{f}_i^s, \mathbf{f}_j^t)], \quad \forall i \in n_c^s, j \in n_c^t : \quad y_{i,c}^s = 1, \ \arg\max_{c'} g_{j,c'}^t = c, \tag{54}$$

where $g_{j,c}^t \in \mathbf{g}_j^t$, and $\mathbf{g}_j^t = \mathscr{G}(\mathbf{f}_j^t)$. Moreover, n_c^s is the number of samples in $\mathscr{D}^s$, where $y_{i,c}^s = 1$. Likewise, n_c^t holds the number of target inputs satisfying $\arg\max_{c'} g_{j,c'}^t = c$.

Here, to enhance robustness against noisy pseudo-labels in the target set, we introduce a confidence weighting scheme derived from a principled, entropy-based measure of prediction uncertainty. The core idea is to quantify the uncertainty of a classifier's output probability vector, $\mathbf{g}_j \in [0,1]^C$, using its Rényi's quadratic entropy in Equation (40), as follows:

$$\hat{H}_2(\mathbf{g}_j^t) = -\log\left(\sum_{c=1}^{C} \left(g_{j,c}^t\right)^2\right). \tag{55}$$

In turn, to create a universally comparable score, this entropy value is normalized by its theoretical maximum, which occurs for a uniform distribution and is equal to $H_{2,\max} = -\log\left(\sum_{c=1}^{C}(1/C)^2\right) = \log(C)$. This yields a normalized uncertainty score $\hat{U}(\mathbf{g}_j^t) = \hat{H}_2(\mathbf{g}_j^t)/\log(C)$, which is bounded in $[0,1]$. Therefore, we propose incorporating a confidence weighting vector $\mathbf{w}^t \in \mathbb{R}^{N_t}$, derived from the normalized uncertainty score $\hat{U}(\mathbf{g}_j^t)$:

$$w_j^t = 1 - \hat{U}(\mathbf{g}_j^t) \tag{56}$$

where $w_j^t \in \mathbf{w}^t$. The latter provides a theoretically grounded mechanism to down-weight ambiguous predictions, a strategy that has proven effective in related contexts for handling label uncertainty [199].

Afterward, a target weighting matrix $\tilde{\mathbf{W}}_t^c \in \mathbb{R}^{n_c^t \times n_c^t}$ can be computed, yielding the following:

$$\tilde{\mathbf{W}}_c^t = \tilde{\mathbf{w}}_c^t (\tilde{\mathbf{w}}_c^t)^\top, \tag{57}$$

where $\tilde{\mathbf{w}}_c^t = \{w_j^t : \arg\max_{c'} g_{j,c'}^t = c\} \in \mathbb{R}^{n_c^t}$.

Now, our CREDA method lies in a novel regularization term that enforces alignment between the class-conditional distributions of the source and target domains. So, we employ a kernel-based quadratic Rényi entropy mutual information estimator (see Section 8.2) and the confidence weighting scheme in Equation (56), as follows:

$$\tilde{\mathrm{I}}_2(\mathbf{K}_c^s; \tilde{\mathbf{K}}_c^t) = \frac{1}{2}\left(H_2(\mathbf{K}_c^s) + H_2(\tilde{\mathbf{K}}_c^t)\right) - H_2(\mathbf{K}_c^{\text{mix}}), \tag{58}$$

where $\tilde{\mathbf{K}}_c^t = \mathbf{K}_c^t \odot \tilde{\mathbf{W}}_c^t$, and

$$\mathbf{K}_c^{\text{mix}} = \begin{pmatrix} \mathbf{K}_c^s & \mathbf{K}_c^{st} \\ (\mathbf{K}_c^{st})^\top & \tilde{\mathbf{K}}_c^t \end{pmatrix}, \tag{59}$$

which enables the computation of our MI estimator in Equation (58) even when the source and target sample sizes differ, namely $n_c^t \neq n_c^s$.

Finally, the complete CREDA loss integrates the standard supervised cross-entropy on labeled source data with our proposed mutual information regularizer, based on the quadratic Rényi entropy formulation, as follows:

$$\mathscr{L}_{\text{CREDA}} = \sum_{i=1}^{N_s} \sum_{c=1}^{C} y_{i,c}^s \log\left(\mathscr{G}(\mathscr{F}(\mathbf{x}_i^s))\right) - \lambda \sum_{c \in C} \tilde{\mathrm{I}}_2(\mathbf{K}_c^s; \tilde{\mathbf{K}}_c^t), \tag{60}$$

where $\lambda \in \mathbb{R}^+$ is a hyperparameter controlling the strength of the domain alignment.

In practice, the computation of the kernel matrices in Equations (52)–(54) in our CREDA loss is performed within each training mini-batch. For a given mini-batch of source and target samples, features are first extracted, and pseudo-labels for the target samples are generated. Subsequently, for each class c, the corresponding feature vectors from the source batch (with ground-truth label c) and the target batch (with pseudo-label c) are filtered. The cross-domain kernel matrix $\mathbf{K}_c^{st}$ is then computed by evaluating the Gaussian kernel between every filtered source feature and every filtered target feature from the batch. The intra-domain matrices, $\mathbf{K}_c^s$ and $\mathbf{K}_t^c$, are computed similarly among the respective filtered features. If a class is not present in a given mini-batch, its contribution to the regularization loss for that training step is zero. This batch-wise, class-conditional procedure allows for an efficient and scalable implementation of our proposed alignment objective.

Remarkably, the selection of Rényi's quadratic entropy ($\alpha = 2$) is motivated by its direct connection to the IP in Equation (42), which, under a Gaussian kernel, translates the alignment objective into a geometrically intuitive goal [192]. Specifically, the sample-based estimator in Equation (44) becomes a sum of pairwise similarities, meaning that minimizing our class-conditional loss in Equation (60) is equivalent to encouraging feature vectors of the same class to form tight, pure clusters in the feature space, directly promoting class separability. Furthermore, our approach is sensitive to higher-order

statistics; thereby, CREDA-based loss captures the overall structure of the distributions, such as their dispersion and modality, which is critical for aligning complex, multi-modal classes often found in real-world datasets. Finally, the estimator's formulation as an average over all pairwise interactions provides a robust estimate of class-wise distributional similarity. This inherent averaging makes the gradient estimates stable by mitigating the influence of individual outliers or noisy pseudo-labels, a common challenge in unsupervised settings.

Moreover, in discussing the convergence properties of our CREDA loss, it is crucial to distinguish between the statistical consistency of the estimator and the empirical convergence of the deep learning model during training. The mutual information estimator in Equation (58) inherits strong theoretical properties from its foundation in Parzen-window kernel estimation (see Equation (41)). As established in non-parametric statistics, KDE provides a consistent estimator, meaning the estimated probability density converges to the true underlying density as the number of samples approaches infinity [194]. Consequently, the IP at the core of our approach, and by extension our full mutual information estimator, are also statistically consistent estimators of the true quadratic Rényi's mutual information between the class-conditional distributions. Now, from an optimization perspective, the complete CREDA loss is non-convex due to the highly nonlinear nature of deep approaches. Therefore, formal guarantees of convergence to a global minimum are not feasible, a common characteristic of deep learning systems. Still, our method is designed to facilitate stable empirical convergence. The use of an infinitely differentiable Gaussian kernel ensures our regularization term is smooth, contributing to a well-behaved loss landscape that is conducive to gradient-based optimization.

Figure 18 summarizes the core components and training pipeline of our proposed CRERDA model for conditional domain adaptation.

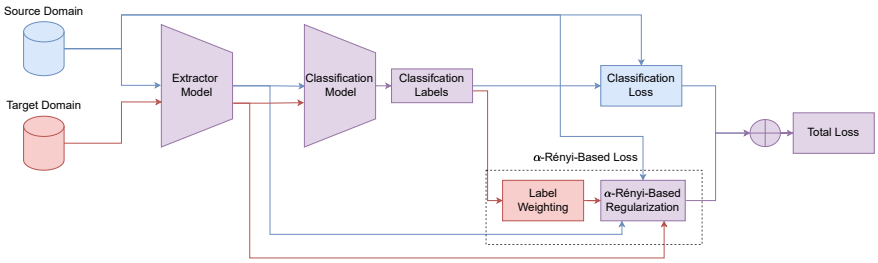


Figure 18: CREDA framework for domain adaptation, incorporating classification loss and α-Rényi Entropy-based label weighting and regularization to attain domain alignment with a class-aware structure. **Blue**: source, **Red**: target, **Purple**: shared.

The development of the comprehensive CREDA methodology detailed above is systematically organized within a structured research timeline. This plan not only chronicles the work completed to date but also charts the course for future research extensions. Specifically, the theoretical formulation of kernel methods and the kernel-based α-Rényi entropy estimator (Sections 8.1 and 8.2) constitutes the core of Objective 1 in this plan. Subsequently, the design and implementation of the complete domain adaptation frame-

work, including the conditional entropy regularization and the label weighting scheme (Section 8.3), correspond to Objective 2. The following schedule provides a detailed account of these phases and outlines the subsequent research activities.

8.4 Implementation Schedule

Table 1 presents a detailed 36-month schedule for the proposed research, structured around the specific objectives and future work. Activities accomplished during the first 18 months, representing the development and initial validation of the CREDA framework, are highlighted in gray.

The subsequent 18 months will focus on advanced research extensions, culminating in at least two key scientific publications:

- An extension of CREDA for robust **cross-subject and cross-dataset EEG classification**, addressing the critical challenge of generalization across different BCI paradigms and hardware.
- The development of an **advanced entropy-based weighting mechanism** specifically designed to handle the highly non-stationary and noisy nature of EEG signals, further improving the framework's stability and performance.

This period also includes a tentative research internship scheduled for 2026 to collaborate with leading experts in the field. These research activities will form the core of the final compilation and writing of the PhD thesis, with this final phase highlighted in blue.

Table 1: 36-month implementation schedule (current progress: **Month 18**).

Months	Objective 1: Entropy Estimator		Objective 2: CREDA Framework		Objective 3: Interpretability		Future Work			Dissemination	
	1a	1b	2a	2b	3a	3b	FW1	FW2	FW3	Papers	Thesis
1–3	✔		✔								
4–6	✔	✔	✔	✔							
7–9		✔		✔							
10–12		✔		✔	✔						
13–15		✔		✔	✔					✔	
16–18		✔		✔	✔					✔	
19–21							✔				
22–24						✔	✔	✔			
25–27						✔		✔	✔	✔	❖
28–30								✔	✔		✔
31–33									✔	✔	✔
34–36											✔

Legend: ✔ = completed ❖ (blue cell) = thesis-writing phase.
Task codes: 1a/2a – theoretical formulation; 1b/2b – model implementation and validation; 3a – basic interpretability; 3b – advanced (EEG) interpretability; FW1 – multi-source/continual DA; FW2 – multidimensional explanations; FW3 – large-scale validation.

9 Experimental Set-Up and Results on Image Data

To rigorously evaluate the effectiveness of the proposed CREDA framework for domain adaptation in image classification tasks, we present a comprehensive analysis that in-

cludes descriptions of the benchmark datasets, training protocols, comparative baselines, and quantitative and qualitative performance assessments.

9.1 Tested Datasets

To assess the effectiveness and robustness of the proposed domain adaptation method, we conducted extensive experiments on three widely recognized benchmark datasets commonly used in domain adaptation research. Each dataset encompasses visual domains exhibiting substantial distribution shifts, thereby providing a challenging setting for learning domain-invariant representations, as detailed below:

– *Digits:* This benchmark suite is designed for evaluating domain adaptation on digit recognition tasks, spanning both handwritten and natural-scene digits. It comprises three standard datasets: MNIST (M), a large database of handwritten digits; USPS (U), another handwritten digit set characterized by its lower resolution; and SVHN (S), which contains house numbers cropped from real-world street-level images [200]. Notably, the S domain is particularly challenging due to its significant variability in lighting, background clutter, and visual styles compared to M and U (see Figure 19).

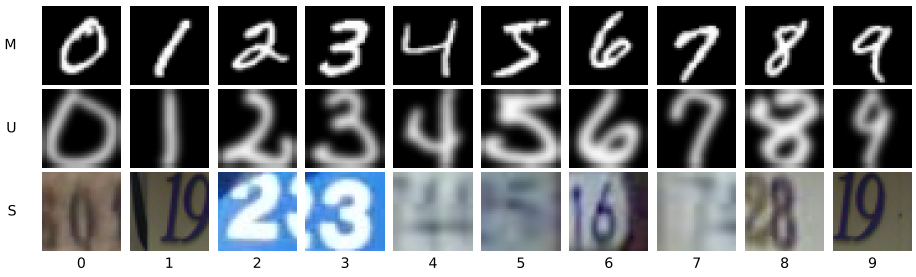


Figure 19: Representative input images for each digit class across source and target domains.

– *ImageCLEF-DA:* This is a standard benchmark for unsupervised domain adaptation, organized as part of the ImageCLEF evaluation campaign. It comprises 12 common object classes shared across three distinct visual domains: Caltech-256 (C), ImageNet ILSVRC 2012 (I), and Pascal VOC 2012 (P), see Figure 20. Each domain contains 600 images, with a balanced distribution of 50 images per class [201]. All images are resized to 224×224 pixels.

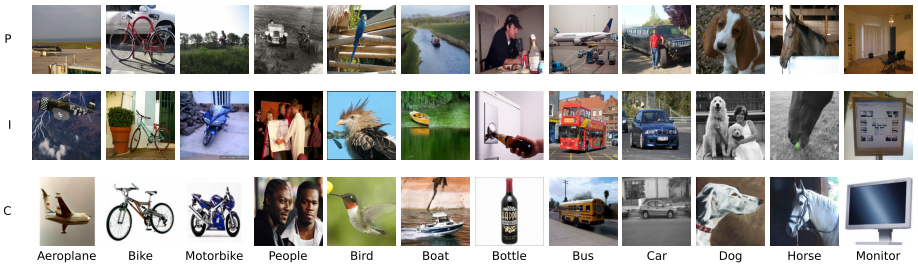


Figure 20: Representative input images for each object class across source and target domains in the ImageCLEF-DA dataset.

– *Office-31:* It consists of 4110 images across 31 object classes, sourced from three domains with distinct visual characteristics: Amazon (A), which features centered objects on a clean, white background under controlled lighting; Webcam (W), containing low-resolution images with typical noise and color artifacts; and DSLR (D), which includes high-resolution images with varying focus and lighting conditions [202]. Here, we selected a subset of ten shared classes (see Figure 21).



Figure 21: Representative input images for each object class across source and target domains in the Office-31 dataset.

Together, these benchmarks allows evaluating the capacity of domain adaptation methods to generalize across diverse and challenging visual domains.

9.2 Assessment and Method Comparison

To comprehensively evaluate the impact of the feature extractor's architecture on model performance, we experimented with three distinct backbones: a standard ResNet-18, its deeper counterpart ResNet-50, and a ViT. Each backbone is adapted for feature extraction in domain transfer tasks by removing its final classification layer. The primary baseline is a ResNet-18 convolutional backbone pretrained on ImageNet [203]. To tailor the architecture for our tasks, the final fully connected layer is removed, while all preceding convolutional and residual blocks are retained. This modification enables

the extraction of high-level spatial representations that are robust and transferable across domains [204]. A comprehensive description of the ResNet-18 feature extractor's architecture is provided in Table 2.

Afterward, to investigate the effect of network depth, we also employed a ResNet-50 backbone, a deeper and more powerful variant within the ResNet family [203]. ResNet-50 utilizes bottleneck residual blocks, which are more computationally efficient for deeper networks [205]. Similar to the ResNet-18 configuration, the model is pretrained on ImageNet, and its final fully connected layer is removed to serve as a feature extractor. This results in a 2048-dimensional feature vector. The detailed architecture is presented in Table 3.

Table 2: Architectural details of the ResNet-18 feature extractor.

Layer Name	**Type**	**Input Shape**	**Output Shape**	**Param. #**
Input	InputLayer	$(3, \check{H}, \check{W})$	$(3, \check{H}, \check{W})$	0
Conv1	Conv2D + BN + ReLU	$(3, \check{H}, \check{W})$	$(64, \check{H}/2, \check{W}/2)$	9408
MaxPool	MaxPooling	$(64, \check{H}/2, \check{W}/2)$	$(64, \check{H}/4, \check{W}/4)$	0
Layer1	Residual Block $\times$ 2	$(64, \check{H}/4, \check{W}/4)$	$(64, \check{H}/4, \check{W}/4)$	73,728
Layer2	Residual Block $\times$ 2	$(64, \check{H}/4, \check{W}/4)$	$(128, \check{H}/8, \check{W}/8)$	230,144
Layer3	Residual Block $\times$ 2	$(128, \check{H}/8, \check{W}/8)$	$(256, \check{H}/16, \check{W}/16)$	919,040
Layer4	Residual Block $\times$ 2	$(256, \check{H}/16, \check{W}/16)$	$(512, \check{H}/32, \check{W}/32)$	3,674,112
AvgPool	GlobalAvgPooling	$(512, \check{H}/32, \check{W}/32)$	$(512, 1, 1)$	0
Flatten	Flatten	$(512, 1, 1)$	(512)	0

Table 3: Architectural details of the ResNet-50 feature extractor.

Layer Name	**Type**	**Input Shape**	**Output Shape**	**Param. #**
Input	InputLayer	$(3, \check{H}, \check{W})$	$(3, \check{H}, \check{W})$	0
Conv1	Conv2D + BN + ReLU	$(3, \check{H}, \check{W})$	$(64, \check{H}/2, \check{W}/2)$	9408
MaxPool	MaxPooling	$(64, \check{H}/2, \check{W}/2)$	$(64, \check{H}/4, \check{W}/4)$	0
Layer1	Bottleneck Block $\times$ 3	$(64, \check{H}/4, \check{W}/4)$	$(256, \check{H}/4, \check{W}/4)$	214,528
Layer2	Bottleneck Block $\times$ 4	$(256, \check{H}/4, \check{W}/4)$	$(512, \check{H}/8, \check{W}/8)$	1,182,720
Layer3	Bottleneck Block $\times$ 6	$(512, \check{H}/8, \check{W}/8)$	$(1024, \check{H}/16, \check{W}/16)$	7,084,032
Layer4	Bottleneck Block $\times$ 3	$(1024, \check{H}/16, \check{W}/16)$	$(2048, \check{H}/32, \check{W}/32)$	15,085,568
AvgPool	GlobalAvgPooling	$(2048, \check{H}/32, \check{W}/32)$	$(2048, 1, 1)$	0
Flatten	Flatten	$(2048, 1, 1)$	(2048)	0

Also, to explore an alternative architectural paradigm beyond convolutional networks, we incorporated a ViT-based model, specifically the `vit_tiny_patch16_224` variant (termed ViT-Tiny) [206]. Unlike CNNs, ViT-Tiny processes images by splitting them into a sequence of fixed-size patches, which are then linearly embedded and fed into a standard Transformer encoder. For this study, we use a ViT-Tiny pretrained on ImageNet with an input resolution of 224×224. The classification head is discarded, and the output embedding of the special `[CLS]` token from the final Transformer block

is used as the feature representation, yielding a 192-dimensional vector. The architecture is detailed in Table 4.

Table 4: Architectural details of the ViT-Tiny feature extractor.

Layer Name	**Type**	**Input Shape**	**Output Shape**	**Param. #**
Input	InputLayer	$(3, 224, 224)$	$(3, 224, 224)$	0
Patch Embedding	Conv2D (Patching)	$(3, 224, 224)$	$(196, 192)$	147,648
Add CLS Token	Concatenation	$(196, 192)$	$(197, 192)$	192
Add Pos. Embedding	Parameter Add	$(197, 192)$	$(197, 192)$	37,824
Transformer Encoder	Encoder Block $\times$ 12	$(197, 192)$	$(197, 192)$	5,529,792
Extract CLS Token	Indexing	$(197, 192)$	$(1, 192)$	0
LayerNorm	LayerNorm	$(1, 192)$	$(1, 192)$	384
Flatten	Flatten	$(1, 192)$	(192)	0

Moreover, the following domain adaptation strategies are considered for comparison:

– Baseline: A thenceforward approach is trained exclusively on the source domain without any adaptation mechanism, see Figure 22. The optimization objective is to minimize the conventional supervised cross-entropy loss, which serves as a lower bound for performance evaluation under domain shift:

$$\mathscr{L}_{\text{Baseline}} = -\sum_{i=1}^{N_s}\sum_{c=1}^{C} y_{i,c}^{s} \log\left(\mathscr{G}(\mathscr{F}(\mathbf{x}_i^s))\right). \tag{61}$$

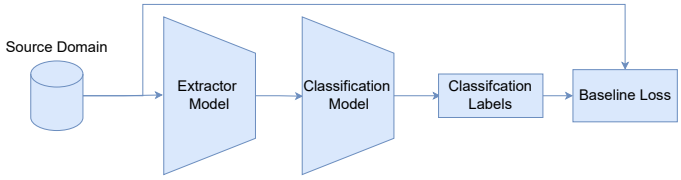


Figure 22: Baseline model for supervised training on the source domain without adaptation.

– DANN: The Domain-Adversarial Neural Network (DANN) [207] introduces a domain discriminator, $\tilde{\mathscr{G}} : \mathbb{R}^d \to [0, 1]$, which is trained to distinguish source features from target ones, see Figure 23. The discriminator is implemented as a multi-layer neural network, where a predicted label of 1 indicates source domain membership, and 0 indicates target domain membership. Moreover, the feature extractor is simultaneously trained to produce features that fool the discriminator, thereby learning domain-invariant representations via a Gradient Reversal Layer (GRL). The overall objective is a minimax game:

$$\mathscr{L}_{\text{DANN}} = \mathscr{L}_{\text{Baseline}} + \check{\lambda}\mathscr{L}_{Adv}, \tag{62}$$

where $\lambda \in \mathbb{R}^+$ represents a trade-off hyperparameter. The domain adversarial loss $\mathscr{L}_{Adv}$ is the binary cross-entropy for domain classification, where source samples are assigned domain label 0, and target samples label 1.

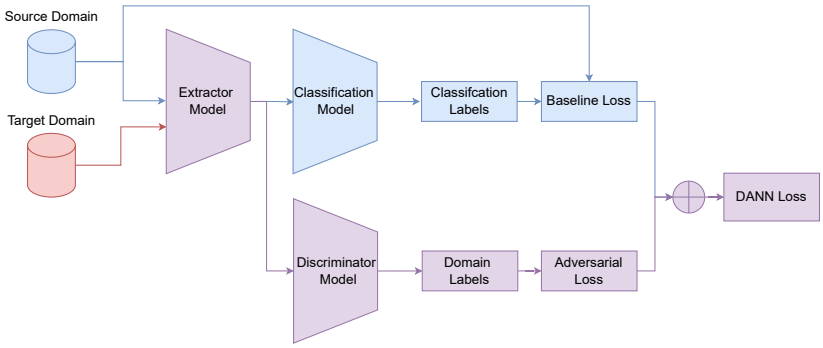


Figure 23: DANN framework for unsupervised domain adaptation. **Blue**: source, **Red**: target, **Purple**: shared.

– ADDA: The Adversarial Discriminative Domain Adaptation (ADDA) framework [208] separates the training into two distinct stages, see Figure 24. First, a source feature extractor $\mathscr{F}^s(\cdot)$ and the classifier $\mathscr{G}(\cdot)$ are trained using the supervised loss $\mathscr{L}_{\text{Baseline}}$ (see Equation (61)). In the second stage, the parameters of $\mathscr{F}^s(\cdot)$ and $\mathscr{G}(\cdot)$ are frozen. Then, a new target feature extractor, $\mathscr{F}^t(\cdot)$ (initialized with the weights in $\mathscr{F}^s(\cdot)$), is then trained to fool the domain discriminator in a minimax game (see Equation (62)). The objective is to align the target feature distribution with the fixed source feature distribution.

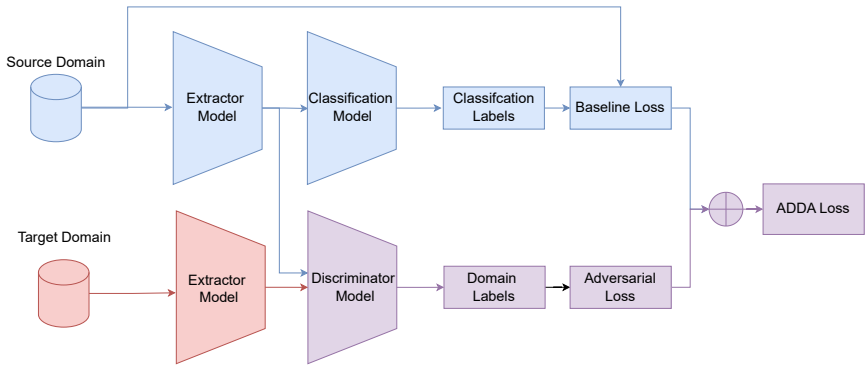


Figure 24: ADDA framework for unsupervised domain adaptation. **Blue**: source; **Red**: target; **Purple**: shared.

– CDAN+E: The Conditional Domain Adversarial Network (CDAN) [209] enhances adversarial alignment by using a multilinear feature representation, $\mathbf{h} =$

$\mathscr{F}(\mathbf{x}) \otimes \mathscr{G}(\mathscr{F}(\mathbf{x}))$, as input to the domain discriminator $\tilde{\mathscr{G}}$. The CDAN+E variant, as implemented in standard benchmarks, employs a sophisticated entropy-based mechanism that serves a dual purpose: it implements entropy minimization for the target domain while simultaneously weighting the adversarial loss to focus on more reliable samples, as seen in Figure 25.

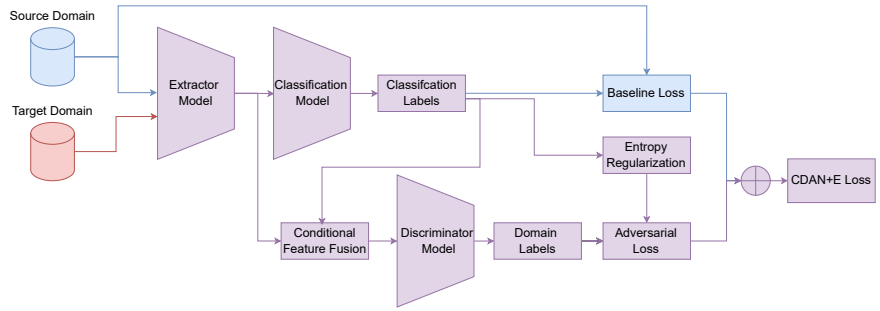


Figure 25: CDAN+E framework for unsupervised domain adaptation. **Blue**: source; **Red**: target; **Purple**: shared.

Specifically, the Shannon entropy $H(\mathbf{g})$ is computed for the predictions $\mathbf{g}$ of all samples in a batch. This entropy value is then used in two ways. First, it is passed through a GRL, which implicitly creates an entropy minimization objective for the feature extractor, encouraging it to produce more confident (low-entropy) predictions. Second, the entropy is transformed into a sample-wise weight, as follows:

$$\breve{w}_j = 1 + \exp(-H(\mathbf{g}_j)). \tag{63}$$

This weighting scheme gives greater importance to samples with confident predictions (low entropy), thereby focusing the adversarial alignment on well-structured regions of the feature space. The resulting weighted conditional adversarial loss, $\mathscr{L}_{Adv}$, is then defined as follows:

$$\mathscr{L}_{Adv} = -\sum_{i=1}^{N_s} \breve{w}_i^s \log\left(\tilde{\mathscr{G}}(\mathbf{h}_i^s)\right) - \sum_{j=1}^{N_t} \breve{w}_j^t \log\left(1 - \tilde{\mathscr{G}}(\mathbf{h}_j^t)\right), \tag{64}$$

where both $\breve{w}_i^s$ and $\breve{w}_j^t$ are calculated according to Equation (63). The total loss for the CDAN+E framework can thus be expressed as the combination of the supervised loss and this integrated adversarial and entropy-regularized objective (see Equation (62)).

Overall, two main components are employed depending on the training objective: a label classifier for supervised task learning and a domain discriminator for adversarial domain adaptation. Namely, the label classifier transforms the feature vector of dimension d, produced by the backbone, into a vector of C class logits. The value of d depends on the specific feature extractor employed (e.g., 512 for ResNet-18, 2048 for ResNet-50, and 192 for ViT-Tiny). The corresponding architecture is presented in Table 5.

Table 5: Architecture of the generic label classifier.

Layer Name	Type	Input Shape	Output Shape	Param. #
Input	InputLayer	$(d,)$	$(d,)$	0
FC1	Dense	$(d,)$	$(d/2,)$	$(d \times d/2) + d/2$
BN1	BatchNorm1d	$(d/2,)$	$(d/2,)$	d
ReLU1	Activation	$(d/2,)$	$(d/2,)$	0
FC2	Dense	$(d/2,)$	$(d/4,)$	$(d/2 \times d/4) + d/4$
BN2	BatchNorm1d	$(d/4,)$	$(d/4,)$	$d/2$
ReLU2	Activation	$(d/4,)$	$(d/4,)$	0
Output	Dense	$(d/4,)$	$(C,)$	$(d/4 \times C) + C$

In adversarial training, a domain discriminator is employed to differentiate between source and target samples, thereby promoting domain-invariant feature extraction. Its input dimension d is determined by the underlying method. For instance, DANN and ADDA use the feature vector directly, while CDAN+E utilizes the outer product between features and class predictions, yielding an input dimension $d \times C$. The architecture, which mirrors the general structure of the label classifier, is detailed in Table 6.

Table 6: Architecture of the generic domain discriminator.

Layer Name	Type	Input Shape	Output Shape	Param. #
Input	InputLayer	$(d,)$	$(d,)$	0
FC1	Dense	$(d,)$	$(d/2,)$	$(d \times d/2) + d/2$
BN1	BatchNorm1d	$(d/2,)$	$(d/2,)$	d
ReLU1	Activation	$(d/2,)$	$(d/2,)$	0
FC2	Dense	$(d/2,)$	$(d/4,)$	$(d/2 \times d/4) + d/4$
BN2	BatchNorm1d	$(d/4,)$	$(d/4,)$	$d/2$
ReLU2	Activation	$(d/4,)$	$(d/4,)$	0
Output	Dense	$(d/4,)$	$(1,)$	$(d/4) + 1$

In all experimental scenarios, we report the classification accuracy and its associated standard deviation in the test set of the target domain. Moreover, during training, model performance is periodically evaluated on validation subsets drawn from both source and target domains to monitor intermediate generalization behavior. In this sense, the Accuracy (ACC) measure is defined as follows:

$$\mathrm{Acc}(\hat{\boldsymbol{y}}, \boldsymbol{y}) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(\hat{y}_i = y_i), \tag{65}$$

where $\hat{y}_i \in \hat{\boldsymbol{y}}$ and $y_i \in \boldsymbol{y}$ denote the predicted and ground truth labels, respectively. $\mathbb{I}(\cdot)$ is the indicator function that returns 1 if the condition is true and 0 otherwise. The standard deviation is estimated from the batch-wise accuracies, serving as a proxy for model stability during inference. The Baseline model is trained solely on labeled samples from the source domain and is directly evaluated in the target domain without any adaptation mechanisms. This setting establishes a lower bound for performance under domain shift conditions.

In addition to quantitative measures, we assess the discriminative quality of the learned feature representations using qualitative techniques. Specifically, we employ the well-known Uniform Manifold Approximation and Projection (UMAP) [182], a nonlinear dimensionality reduction technique to project high-dimensional features into a two-dimensional latent space, enabling visual inspection of inter-domain and inter-class separability [210]. This technique facilitates an empirical evaluation of how well the feature extractor captures semantically consistent structures across domains. To further complement this analysis, we apply the GradCAM++ method to the classifier module in order to visualize spatial attention regions associated with individual predictions [211]. These attention maps provide insight into the decision-making process of the model and support a comparative interpretation of class activation patterns across source and target domains.

9.3 Training Details

The training procedure follows the standard protocol for unsupervised domain adaptation: all labeled data from the source domain are used along with the entire set of unlabeled data from the target domain. The latter approach aims to learn domain-invariant representations without requiring explicit supervision in the target domain.

All models are trained using the Adam optimizer. For the non-adaptive baseline, models are trained with a fixed learning rate (10^{-3} for ResNet architectures and 10^{-4} for ViT-Tiny) and no weight decay. For all domain adaptation methods, a dynamic scheduling scheme is employed for both the learning rate and the adversarial weighting parameter to promote stable convergence and mitigate early overfitting of the discriminator. Both hyperparameters are updated according to the relative training progress $\breve{p} = \frac{\text{epoch}}{\text{total_epochs}}$, according to the following expressions:

$$\eta(\breve{p}) = \eta_0(1+\alpha\breve{p})^{-\beta}, \quad \lambda(\breve{p}) = \frac{1-e^{-\delta\breve{p}}}{1+e^{-\delta\breve{p}}}; \tag{66}$$

where the schedule hyper-hyperparameters are updated to $\alpha = 20$, $\beta = 0.75$, and $\delta = 20$ (see Figure 26).

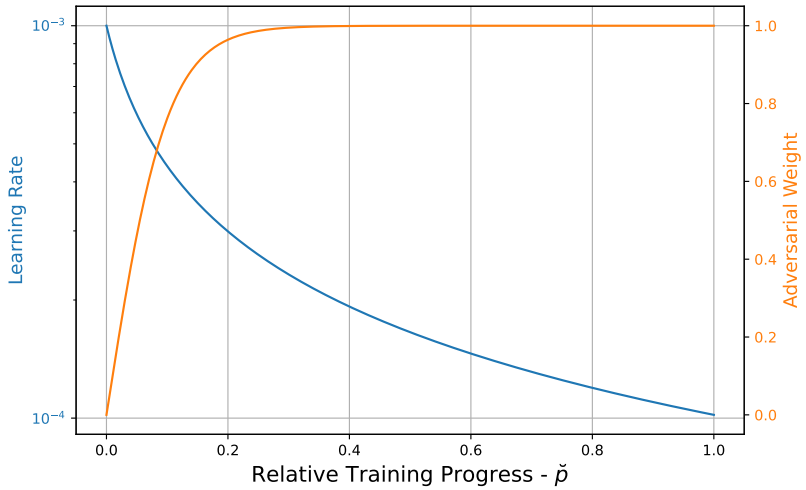


Figure 26: Dynamic scheduling of learning rate and adversarial weighting factor as functions of the relative training progress $\check{p}$ (horizontal axis, dimensionless, 0–1) and their values in logarithmic scale (vertical axis). **Blue:** learning rate $\eta(\check{p})$. **Orange:** adversarial weight $\lambda(\check{p})$, see Equation (66).

In addition to stratified sampling, the batch size is dynamically adjusted based on the size of the training set (N) in each domain, according to the following empirical rule:

$$\text{BatchSize} = \min\left(\max\left(16, \lfloor 0.1 \cdot N \rfloor\right), 64\right). \tag{67}$$

The initial learning rate η_0 was empirically tuned for each model, method, and dataset, typically ranging from 10^{-3} to 10^{-5}. Notably, the first stage of ADDA was trained with a fixed learning rate of 10^{-4}. Furthermore, to adapt the pretrained ViT-Tiny architecture for the lower-resolution Digits dataset (32×32), we applied bicubic interpolation to its positional embeddings. This step was necessary to align the spatial dimensions of the pretrained weights (originally for 224×224 inputs) with the target image size, enabling effective knowledge transfer.

Next, to maintain class balance during model training and evaluation, an initial partition is performed into training (70%), validation (15%), and test (15%) subsets. This process is conducted independently for both the source and target domains. To ensure representative subsets, stratified sampling is applied within each partition, preserving the internal class distributions of each domain. In particular, the independent construction of the validation sets enables consistent and comparable evaluation conditions across domains, which is essential in domain adaptation scenarios where distributional shifts may introduce evaluation bias.

The lower and upper bounds were established empirically. The lower bound ensures the existence of at least 10 mini-batches per epoch, contributing to optimization stability and preventing prohibitively long training times on small datasets. Conversely, the upper bound avoids excessively large batches that could destabilize learning or exceed GPU

memory capacity. This configuration strikes an effective trade-off between gradient stability and computational efficiency, especially when handling domains of different sizes.

It is important to note that, since both dataset partitioning and batch size are determined by the number of available samples in each domain, the number of training instances per epoch is not the same across domains. This asymmetry reflects the inherent scale differences between datasets and allows each domain to contribute proportionally to the learning process without enforcing artificial uniformity.

For all experiments, the kernel bandwidth parameter σ used in the estimation of Rényi's quadratic entropy was adaptively determined for each training batch using the median heuristic. This common practice involves setting σ as the square root of the median of all pairwise squared Euclidean distances within the combined source and target feature batch, as follows [212]:

$$\sigma = \sqrt{\text{median}\left(\left\{\|\mathbf{f}_i^s - \mathbf{f}_j^t\|_2^2\right\}_{i,j}\right) + \varepsilon}, \quad \mathbf{f}_i^s, \mathbf{f}_j^t \in \mathbb{R}^d. \tag{68}$$

This data-driven approach automates a critical hyperparameter, ensuring that the kernel's scale is appropriately tailored to the feature distribution, which enhances the stability and effectiveness of the alignment process across domains.

Moreover, to qualitatively assess the discriminative capacity of the learned features, we apply dimensionality reduction using UMAP, leveraging the GPU-accelerated `cuML` implementation. Unless otherwise stated, the default parameters are set as follows: `n_components` = 2, `n_neighbors` = 80, and `random_state` = 42. Prior to projection, features are normalized with `MinMaxScaler`, which facilitates visual inspection of inter-class and inter-domain separability in the latent space. Also, we employ the GradCAM++ technique via the `torchcam` library to visualize class-specific attention regions within the input images. Representative samples for each class are selected from both source and target domains, and the last convolutional layer of the feature extractor is designated as the target layer. The resulting attention masks are normalized and overlaid on the corresponding images, offering a qualitative perspective on the spatial focus of the model during classification.

Our experiments were conducted on the Google Colab platform, leveraging a high-performance instance equipped with a NVIDIA (Santa Clara, CA, USA) A100 GPU (40.0 GB of VRAM), 83.5 GB of system RAM, and 235.7 GB of disk storage. For full reproducibility, we set a global random seed of 42 across Python, NumPy 2.0.2, and PyTorch (for both CPU and CUDA) and configured the cuDNN backend to use deterministic algorithms, ensuring consistent results from GPU computations. The development environment was based on `Python 3.11.11`, using `PyTorch 2.1.2` for model training, `cuML 25.02.01` for GPU-accelerated UMAP visualization, and `torchcam 0.4.0` for `GradCAM++`. All source code and datasets are publicly available at: `https://github.com/Daprosero/Domain_Adaptation` (accessed on 4 July 2025).

9.4 Results and Discussion

9.4.1 Domain Adaption Results

A fundamental objective in domain adaptation is to learn representations that remain invariant under distributional shifts between domains, commonly referred to as covariate shift. A model's ability to mitigate this challenge is directly reflected in its accuracy on the target domain. To evaluate CREDA's performance quantitatively, we conducted experiments on three widely adopted benchmark datasets using various backbone architectures.

In the digit adaptation tasks (see Table 7), CREDA demonstrates state-of-the-art performance, achieving the highest average accuracy with both ResNet-18 (62.65%) and ResNet-50 (64.07%) backbones. It performs exceptionally well in challenging tasks such as M→U (achieving up to 91.77% with ResNet-50), characterized by significant visual disparities. A noteworthy observation arises with the ViT-Tiny backbone. Here, conventional adversarial methods like DANN, ADDA, and CDAN+E experience a significant performance collapse, falling well below the source-only Baseline. This suggests that the dynamics of adversarial training may be unstable or less compatible with the global, patch-based feature space learned by Transformers, in contrast to the hierarchical features of CNNs. Nevertheless, CREDA is markedly less affected by this architectural shift. While the Baseline achieves the top rank in this specific instance, CREDA's performance (47.23%) remains highly competitive and substantially surpasses other adaptation methods, highlighting its greater architectural robustness.

Table 7: Accuracy (%) on Digits for unsupervised domain adaptation using different backbone architectures.

Model	Method	M → U	M → S	U → M	U → S	S → M	S → U	Avg
ResNet-18	Baseline	56.73 ± 20.93	22.04 ± 14.77	76.64 ± 15.49	9.68 ± 10.60	69.49 ± 16.82	**74.69 ± 15.88**	51.55 ± 15.75
	DANN	86.20 ± 10.64	19.28 ± 13.61	80.84 ± 13.65	**28.65 ± 16.12**	72.64 ± 15.72	70.66 ± 16.33	59.71 ± 14.35
	ADDA	7.68 ± 9.13	30.99 ± 16.93	83.30 ± 12.67	28.27 ± 16.22	**74.32 ± 15.31**	66.27 ± 15.85	54.75 ± 13.49
	CDAN+E	81.99 ± 11.92	15.08 ± 12.80	25.12 ± 15.75	14.19 ± 12.72	56.42 ± 17.01	66.45 ± 17.51	43.21 ± 14.62
	CREDA (ours)	**88.39 ± 11.20**	**32.92 ± 17.14**	**86.08 ± 12.08**	26.29 ± 15.79	71.09 ± 16.01	71.12 ± 15.95	**62.65 ± 14.69**
ResNet-50	Baseline	84.45 ± 14.08	19.61 ± 14.37	64.97 ± 17.51	7.99 ± 9.56	12.91 ± 11.83	66.88 ± 18.16	42.80 ± 14.25
	DANN	90.77 ± 10.20	36.38 ± 16.90	**90.64 ± 10.38**	**21.01 ± 14.46**	73.10 ± 15.59	69.65 ± 15.92	63.59 ± 13.91
	ADDA	84.00 ± 11.52	11.66 ± 11.31	39.03 ± 17.22	14.72 ± 12.28	61.32 ± 16.72	63.16 ± 15.78	45.65 ± 14.14
	CDAN+E	54.20 ± 17.52	17.33 ± 13.40	12.73 ± 12.04	10.96 ± 10.74	30.69 ± 16.69	43.88 ± 19.33	28.30 ± 14.95
	CREDA (ours)	**91.77 ± 9.27**	**37.36 ± 17.78**	80.84 ± 14.01	20.52 ± 14.15	**76.16 ± 14.93**	**77.79 ± 13.54**	**64.07 ± 13.95**
ViT-Tiny	Baseline	67.56 ± 18.44	**27.38 ± 16.10**	**69.82 ± 17.17**	14.11 ± 12.63	**66.28 ± 17.13**	62.50 ± 18.11	**51.28 ± 16.60**
	DANN	26.97 ± 15.90	7.87 ± 9.69	24.84 ± 15.65	9.53 ± 10.36	7.67 ± 9.44	17.28 ± 13.45	15.69 ± 12.42
	ADDA	8.96 ± 10.40	10.19 ± 10.79	13.28 ± 12.51	11.19 ± 11.18	10.08 ± 10.57	2.83 ± 5.67	9.42 ± 10.19
	CDAN+E	16.36 ± 14.27	10.17 ± 10.80	9.74 ± 10.50	9.40 ± 10.30	9.74 ± 10.50	9.14 ± 10.67	10.76 ± 11.17
	CREDA (ours)	**75.69 ± 15.44**	21.51 ± 14.37	47.23 ± 17.26	**17.56 ± 13.50**	54.57 ± 17.67	**66.82 ± 17.84**	47.23 ± 16.01

Similarly, on the ImageCLEF-DA dataset (see Table 8), CREDA's superiority is even more pronounced. It consistently achieves top-tier results, securing the highest average accuracy across all three backbones. Critically, with the ViT-Tiny backbone, CREDA (82.41%) is the only adaptation method to decisively outperform the strong Baseline model (80.19%). This again contrasts sharply with other adversarial methods, which either lag or perform on par with the Baseline. This reinforces the hypothesis that CREDA's Rényi entropy-based regularization offers a more stable and effective path to domain alignment than the adversarial objectives of its counterparts, particularly

when paired with Transformer architectures. These results suggest that our method more effectively balances domain alignment and the preservation of class discriminability.

Table 8: Accuracy (%) on ImageCLEF-DA for unsupervised domain adaptation using different backbone architectures.

Model	Method	I → P	I → C	P → I	P → C	C → I	C → P	Avg
ResNet-18	Baseline	58.00 ± 21.71	76.83 ± 21.52	68.00 ± 19.30	76.50 ± 19.05	49.83 ± 24.44	38.67 ± 25.60	61.31 ± 21.94
	DANN	60.00 ± 25.00	**85.56 ± 13.55**	66.67 ± 16.43	78.89 ± 18.04	**76.67 ± 17.78**	57.78 ± 23.74	70.93 ± 19.09
	ADDA	**68.89 ± 20.87**	77.78 ± 11.10	71.11 ± 19.09	**81.11 ± 16.39**	74.44 ± 14.56	58.89 ± 21.62	72.04 ± 17.27
	CDAN+E	56.67 ± 23.31	62.22 ± 15.50	65.56 ± 16.71	58.89 ± 16.28	68.89 ± 21.62	47.78 ± 22.90	60.00 ± 19.39
	CREDA (ours)	66.67 ± 19.31	82.22 ± 17.24	**77.78 ± 16.28**	80.00 ± 18.56	73.33 ± 19.22	**62.22 ± 18.81**	**73.70 ± 18.24**
ResNet-50	Baseline	46.50 ± 26.20	56.83 ± 33.03	69.67 ± 20.05	80.17 ± 16.71	55.50 ± 20.47	43.00 ± 23.72	58.61 ± 23.36
	DANN	**77.78 ± 17.13**	90.00 ± 9.42	**82.22 ± 13.41**	86.67 ± 16.10	**86.67 ± 10.66**	**75.56 ± 11.72**	**83.15 ± 13.07**
	ADDA	80.00 ± 14.60	80.00 ± 18.84	82.22 ± 17.24	82.22 ± 18.04	85.56 ± 11.25	73.33 ± 13.06	80.56 ± 15.51
	CDAN+E	48.89 ± 25.19	60.00 ± 23.31	58.89 ± 14.43	52.22 ± 24.11	56.67 ± 13.59	42.22 ± 23.13	53.15 ± 20.63
	CREDA (ours)	72.22 ± 18.04	**92.22 ± 9.91**	**82.22 ± 13.41**	**90.00 ± 12.07**	85.56 ± 12.45	67.78 ± 13.41	81.67 ± 13.22
ViT-Tiny	Baseline	73.00 ± 24.32	92.50 ± 11.44	**85.67 ± 15.47**	**88.00 ± 12.57**	77.17 ± 25.62	64.83 ± 29.69	80.19 ± 19.85
	DANN	70.00 ± 22.06	91.11 ± 6.15	81.11 ± 15.50	86.67 ± 11.92	83.33 ± 16.96	62.22 ± 15.50	79.07 ± 14.68
	ADDA	68.89 ± 18.04	90.00 ± 7.77	78.89 ± 12.45	85.56 ± 12.45	7.78 ± 9.91	48.89 ± 22.19	63.33 ± 13.80
	CDAN+E	70.00 ± 18.07	87.78 ± 9.91	72.22 ± 15.50	78.89 ± 15.84	36.67 ± 20.64	24.44 ± 13.93	61.67 ± 15.65
	CREDA (ours)	**77.78 ± 18.72**	**93.33 ± 8.43**	80.00 ± 13.59	86.67 ± 15.19	**87.78 ± 11.25**	**68.89 ± 18.04**	**82.41 ± 14.20**

Lastly, on the Office-31 benchmark (see Table 9), CREDA confirms its superiority by achieving the highest average accuracy across all backbones, peaking at 92.96% with ResNet-50. The trend of architectural robustness continues, as CREDA again outperforms all other methods with ViT-Tiny, achieving an average accuracy of 89.31% against the Baseline's 85.46%. The fragility of other methods is particularly evident here, with DANN, ADDA, and CDAN+E suffering catastrophic performance drops (e.g., 20.14% for DANN on D $\rightarrow$ A), rendering them less effective than a simple no-adaptation approach. This consistently demonstrates CREDA's ability not only to adapt effectively but also to generalize its mechanism across fundamentally different architectural paradigms, from convolutional to attention-based models.

To provide a robust statistical assessment of our method's consistency and superiority, we conducted a Friedman test on the accuracy ranks across all nine experimental configurations (three datasets $\times$ three backbones). The test revealed a statistically significant difference among the methods' performances ($\chi^2(4) = 23.21$, $p < 1.15 \times 10^{-4}$), thus allowing us to reject the null hypothesis that all approaches perform equally. This result provides strong evidence that the observed differences in performance are not due to random chance.

Table 10 shows that CREDA achieves the best (lowest) average rank of 1.22. Furthermore, its performance stability is underscored by a remarkably low standard deviation (± 0.44), the lowest among all evaluated methods. This indicates that CREDA consistently ranked at or near the top, irrespective of the dataset or backbone architecture. In contrast, methods like the Baseline (3.44 ± 1.42) exhibit much higher variance, suggesting their performance is less stable across different settings. This statistical validation robustly confirms that CREDA's leading performance is not an artifact of specific experimental conditions but rather a consistent and significant advantage across a diverse range of domains and model architectures, including the challenging Transformer-based setups where other adaptation techniques falter.

Table 9: Accuracy (%) on Office-31 for unsupervised domain adaptation using different backbone architectures.

Model	Method	A → W	A → D	W → A	W → D	D → A	D → W	Avg
ResNet-18	Baseline	50.51 ± 29.45	55.41 ± 25.37	54.91 ± 34.50	96.82 ± 7.98	46.56 ± 34.31	78.98 ± 28.90	63.86 ± 26.75
	DANN	73.33 ± 15.81	87.50 ± 12.50	67.36 ± 16.68	**100.00 ± 0.00**	51.39 ± 17.09	84.44 ± 12.29	77.34 ± 12.40
	ADDA	62.22 ± 26.71	87.50 ± 12.50	54.17 ± 19.65	**100.00 ± 0.00**	59.72 ± 17.45	84.44 ± 9.41	74.68 ± 14.29
	CDAN+E	64.44 ± 14.72	75.00 ± 12.50	50.69 ± 21.64	**100.00 ± 0.00**	51.39 ± 16.54	88.89 ± 12.29	71.74 ± 12.95
	CREDA (ours)	**82.22 ± 18.82**	**91.67 ± 14.43**	**74.31 ± 17.40**	**100.00 ± 0.00**	**74.31 ± 17.40**	**93.33 ± 10.46**	**85.97 ± 13.09**
ResNet-50	Baseline	48.14 ± 29.65	46.50 ± 31.67	53.44 ± 29.99	76.43 ± 24.05	53.86 ± 36.86	90.51 ± 21.12	61.48 ± 28.89
	DANN	**88.89 ± 9.41**	95.83 ± 7.22	**91.67 ± 8.57**	**100.00 ± 0.00**	81.94 ± 13.71	91.11 ± 10.21	91.57 ± 8.19
	ADDA	**88.89 ± 9.41**	95.83 ± 7.22	88.89 ± 9.48	**100.00 ± 0.00**	84.72 ± 13.93	91.11 ± 10.21	91.57 ± 8.37
	CDAN+E	73.33 ± 15.81	75.00 ± 12.50	70.14 ± 19.71	95.83 ± 7.22	52.08 ± 15.01	68.89 ± 20.41	72.55 ± 15.11
	CREDA (ours)	86.67 ± 11.18	**100.00 ± 0.00**	**91.67 ± 8.57**	**100.00 ± 0.00**	**86.11 ± 12.04**	**93.33 ± 10.46**	**92.96 ± 7.04**
ViT-Tiny	Baseline	85.42 ± 21.61	86.62 ± 14.96	80.27 ± 22.52	**100.00 ± 0.00**	66.18 ± 28.50	**94.24 ± 12.00**	85.46 ± 16.60
	DANN	53.33 ± 18.62	75.00 ± 25.00	19.44 ± 12.29	91.67 ± 7.22	20.14 ± 12.23	91.11 ± 11.23	58.45 ± 14.43
	ADDA	80.00 ± 15.31	**95.83 ± 7.22**	12.50 ± 10.50	**100.00 ± 0.00**	15.28 ± 13.93	55.56 ± 16.61	59.86 ± 10.60
	CDAN+E	17.78 ± 24.58	62.50 ± 12.50	20.14 ± 12.96	**100.00 ± 0.00**	21.53 ± 12.72	8.89 ± 10.21	38.47 ± 12.16
	CREDA (ours)	**93.33 ± 6.85**	91.67 ± 7.22	**88.19 ± 16.86**	**100.00 ± 0.00**	**71.53 ± 16.50**	91.11 ± 20.41	**89.31 ± 11.31**

Table 10: Average classification rank of all methods across datasets and model architectures. Ranks are assigned per block (row) based on average accuracy. The final row presents the mean rank ± standard deviation for each method. The Friedman test confirms a significant difference in performance ($\chi^2 = 23.21$, $p < 1.15 \times 10^{-4}$).

Dataset	Backbone	Baseline	DANN	ADDA	CDAN+E	CREDA (Ours)
Digits	ResNet-18	4.0	2.0	3.0	5.0	1.0
	ResNet-50	4.0	2.0	3.0	5.0	1.0
	ViT-Tiny	1.0	3.0	5.0	4.0	2.0
ImageCLEF-DA	ResNet-18	4.0	3.0	2.0	5.0	1.0
	ResNet-50	4.0	1.0	3.0	5.0	2.0
	ViT-Tiny	2.0	3.0	4.0	5.0	1.0
Office-31	ResNet-18	5.0	2.0	3.0	4.0	1.0
	ResNet-50	5.0	2.0	3.0	4.0	1.0
	ViT-Tiny	2.0	4.0	3.0	5.0	1.0
Mean Rank ± Std	–	3.4 ± 1.4	2.4 ± 0.9	3.2 ± 0.8	4.7 ± 0.5	**1.2 ± 0.4**

9.4.2 Interpretability Results

To clarify the reasons for these performance disparities, it is crucial to first examine the inherent complexity of the data domains. Figure 27 presents the 2D UMAP projections of the original feature space, visualized independently for each domain. These plots reveal a fundamental challenge that extends beyond domain shift: the limited class separability within individual domains. This limitation is particularly pronounced in complex datasets such as ImageCLEF-DA and Office-31, where class instances (depicted by distinct colors) exhibit significant entanglement, forming dense and unstructured distributions. Such inherent visual similarity among categories not only complicates classification within the source domain but also serves as a principal source of noisy

pseudo-labels in the target domain during unsupervised adaptation. Consequently, a robust domain adaptation strategy must not only align cross-domain distributions but also construct feature representations that enhance inter-class discrimination.

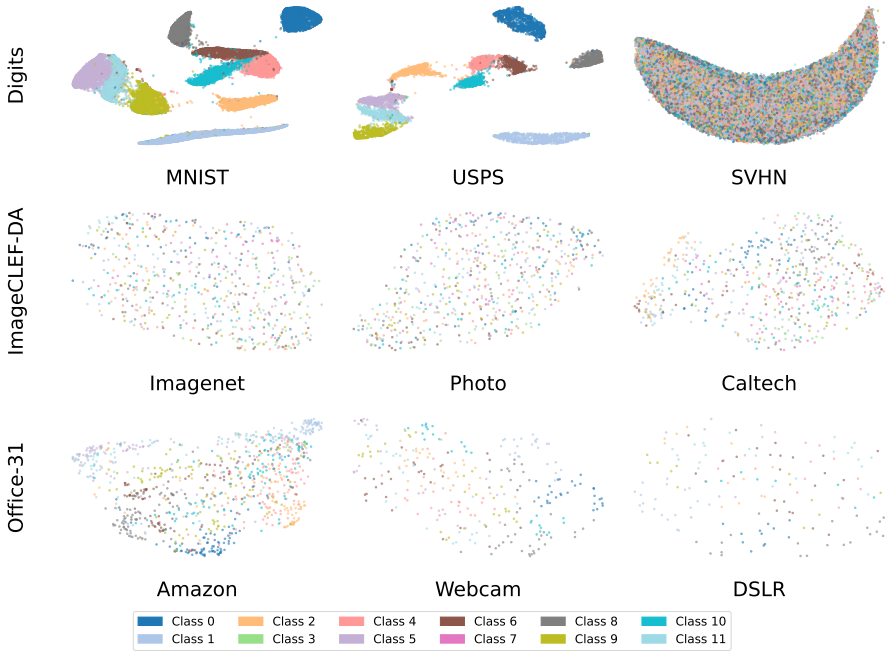


Figure 27: Two-dimensional UMAP projections of original feature representations before domain adaptation. **Rows:** evaluated benchmarks. **Columns:** domains within each benchmark.

Building upon this analysis, Figure 28 illustrates how different adaptation techniques address these structural challenges, visualized through UMAP projections of the learned latent spaces. The first column depicts the initial state prior to training, highlighting both the pronounced domain gap (e.g., M $\rightarrow$ U) and the poor semantic organization (e.g., I $\rightarrow$ C). The Baseline model, trained exclusively on source data, fails to bridge this gap, maintaining a clear division between domains. In contrast, adversarial methods like DANN and ADDA achieve some domain alignment, but often at the expense of class coherence, resulting in fragmented (as seen in M $\rightarrow$ U) and disordered representations across all tasks. While CDAN+E introduces a modest improvement in structural consistency, significant inter-class dispersion remains. Ultimately, CREDA yields a markedly superior configuration: it not only facilitates seamless domain integration—evidenced by the homogeneous blending of source and target samples—but also preserves (M $\rightarrow$ U) and, notably, enhances (I $\rightarrow$ C, W $\rightarrow$ D) class-wise separability, as demonstrated by the emergence of compact, well-defined clusters from initially unstructured feature spaces. This outcome provides a visual explanation for CREDA's superior quantitative performance, indicating its ability to balance the removal of spurious domain-specific

cues with the preservation and recovery of underlying semantic structure.

Having established CREDA's capacity to address covariate shift, we next assess whether the learned representations preserve semantic coherence under concept shift, where object appearance changes substantially across domains. In this context, Figure 29 presents UMAP projections with embedded images to qualitatively examine the model's ability to cluster semantically related concepts.

The results indicate that CREDA learns a semantically rich feature space that transcends superficial variability. For instance, in the M $\rightarrow$ U task, the model accurately groups digits despite substantial stylistic differences, as seen in the clusters corresponding to digits 6, 0, and 4. In the I $\rightarrow$ C task, it successfully groups semantically similar but visually diverse objects, forming distinct clusters for categories like airplanes and bottles despite variations in perspective and background. Similarly, in the W $\rightarrow$ D task, objects such as keyboards and mugs are grouped according to their semantic identity, overcoming differences in image quality. Altogether, these visualizations demonstrate that CREDA not only aligns domains but also constructs a feature space in which proximity reflects conceptual similarity—an essential attribute for robust generalization in real-world applications.

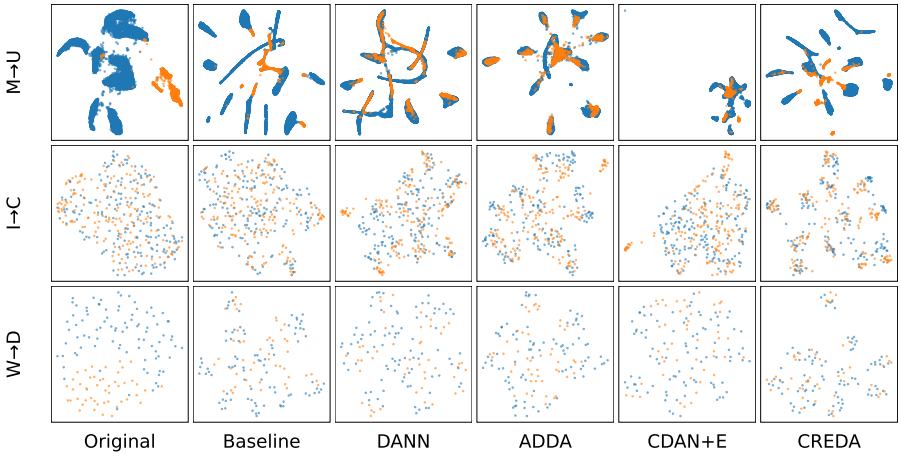


Figure 28: UMAP projections of the learned feature representations across domain adaptation methods, with the source domain shown in blue and the target domain in orange. **Rows:** datasets used in the evaluation. **Columns:** compared adaptation models.



Figure 29: UMAP projections of learned feature representations under the CREDA model, with input images overlaid, where source domain samples appear in blue and target domain samples in orange. **Left**: Digits. **Middle**: ImageCLEF-DA. **Right**: Office-31.

Finally, to reinforce the model's reliability, it is essential not only to demonstrate high accuracy and semantic coherence, but also to ensure that its predictions are grounded in interpretable reasoning. In other words, it must be verified that decisions are driven by relevant visual cues rather than spurious correlations.

To address this, we employ Grad-CAM++, with the results shown in Figure 30. The heatmaps reveal strong semantic consistency: regardless of the domain, the model focuses attention on canonical and representative regions of the object, such as the face in a portrait or the main structural components of a vehicle. This confirms that CREDA does not rely on superficial distribution alignment, but rather performs deep and meaningful semantic knowledge transfer. These findings not only enhance trust in the model's predictions but also establish CREDA as a transparent and robust solution for domain adaptation, strengthening the interpretability and reliability of its outputs.



Figure 30: Class-wise visual explanations under the CREDA model. Each pair of images shows the source domain on the **left** and the corresponding target domain on the **right**. Heatmaps highlight the most salient regions contributing to the predicted class.

9.4.3 Training and Inference Time Analysis

To assess practical viability, we measured training and inference time on the M $\rightarrow$ S task, selected for being the most extensive dataset combination. ResNet-50 was used as the feature extractor due to its higher computational demand relative to the other backbones, offering a conservative estimate of resource requirements. As shown in Figure 31, ADDA incurs the highest training cost due to its two-phase architecture, whereas single-stage methods (DANN, CDAN+E, and CREDA) introduce only a marginal overhead compared to the Baseline. Regarding inference, all adapted models were highly efficient. Notably, DANN, CDAN+E, and CREDA demonstrated slightly faster inference than the Baseline and ADDA, potentially because the learned domain-invariant features streamline the forward pass. This analysis confirms that CREDA offers a compelling trade-off, delivering superior accuracy with a manageable training cost while maintaining efficient inference speeds suitable for real-world deployment.

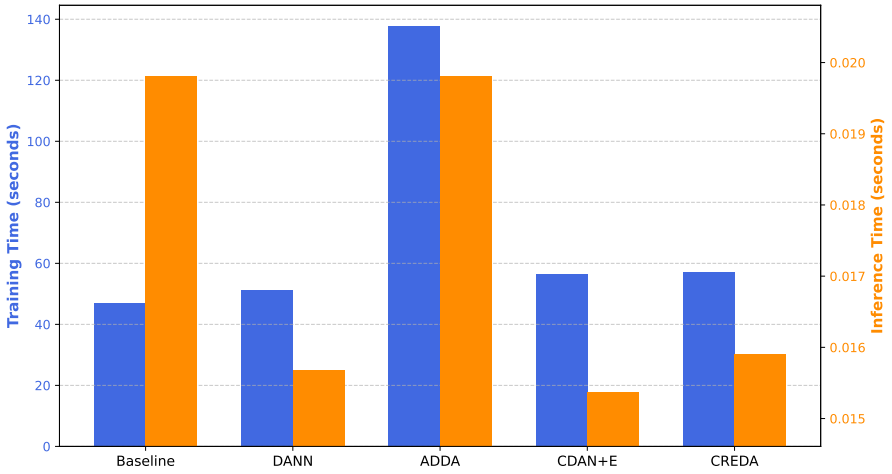


Figure 31: Training and inference time comparison across domain adaptation methods. The left axis shows training time per epoch, while the right axis shows average inference time per sample.

10 Experimental Set-Up and Results on EEG Data

10.1 Tested Datasets

The **DB-I (GiGaScience)** dataset (`http://gigadb.org/dataset/100295`) comprises electroencephalographic (EEG) recordings from 52 participants, of whom 50 were retained for analysis based on signal quality. Each subject completed a single motor imagery (MI) session, structured into five to six blocks with approximately 100 to 120 trials per class (i.e., left- and right-hand imagery).

Each trial followed a fixed temporal structure lasting seven seconds: a two-second resting period with a blank screen, followed by a three-second visual cue instructing the subject to perform the MI task, and ending with a final two-second blank screen. A variable inter-trial interval (ITI) ranging from 0.1 to 0.8 seconds was inserted between trials.

EEG signals were recorded using a 64-channel montage based on the international 10-10 system. Electrodes were arranged beginning in the left frontal region, progressing posteriorly in a serpentine fashion, and returning anteriorly along the sagittal midline to the CPZ position. The sampling rate was 512 Hz. In addition to motor imagery data, the dataset includes recordings of executed movements and six non-task-related artifact conditions (e.g., eye movements, jaw clenching, and resting state).

Figure 32 provides a schematic summary of the trial timeline and electrode configuration used during acquisition.

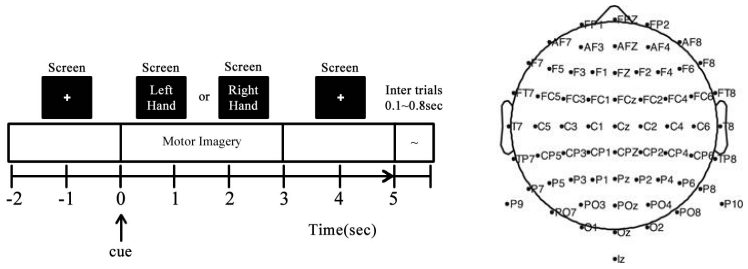


Figure 32: Schematic representation of the MI-EEG experimental protocol. Left: timeline of a single trial, including the initial rest period, visual cue, and post-task interval. Right: spatial arrangement of the EEG electrodes following the 10-10 system.

10.2 Deep Learning Architectures

The second family of experiments employs two compact yet expressive backbones tailored to EEG analysis. The first is a variant of *EEGNet* [213], expressly designed for multichannel biosignals. It comprises (i) a temporal convolution that functions as a learnable band-pass filter, (ii) a depthwise spatial convolution that models inter-channel dependencies, and (iii) a separable convolutional block that fuses spatial and temporal information while preserving architectural efficiency. Table 11 details the instantiated configuration for signals with 64 channels and a temporal length of 320 samples.

Table 11: EEGNet architecture in symbolic form for 64-channel, 320-sample signals.

Layer name	**Type**	**Input shape**	**Output shape**	**Param.#**
Input	InputLayer	(1, 64, 320)	(1, 64, 320)	0
TemporalConv	Conv2D	(1, 64, 320)	(32, 64, 320)	2,048
BN1	BatchNormalization	(32, 64, 320)	(32, 64, 320)	64
SpatialConv	DepthwiseConv2D	(32, 64, 320)	(64, 1, 320)	4,096
BN2	BatchNormalization	(64, 1, 320)	(64, 1, 320)	128
ELU1	Activation	(64, 1, 320)	(64, 1, 320)	0
AvgPool1	AveragePooling2D	(64, 1, 320)	(64, 1, 20)	0
Dropout1	Dropout	(64, 1, 20)	(64, 1, 20)	0
SepConv	SeparableConv2D	(64, 1, 20)	(64, 1, 20)	8,192
BN3	BatchNormalization	(64, 1, 20)	(64, 1, 20)	128
ELU2	Activation	(64, 1, 20)	(64, 1, 20)	0
AvgPool2	AveragePooling2D	(64, 1, 20)	(64, 1, 1)	0
Dropout2	Dropout	(64, 1, 1)	(64, 1, 1)	0
Flatten	Flatten	(64, 1, 1)	(64)	0

This configuration is fixed across all experiments: the temporal stage uses F_1=32 filters, the separable stage uses F_2=64 filters, and the depthwise spatial stage adopts an expansion factor $\tilde{D}$=2. Temporal resolution is reduced via two average-pooling operations with factor 16 each. All convolutions are bias-free, and 2D dropout operates at a fixed rate of 0.2.

The second backbone is *ShallowConvNet*, conceptually aligned with the Filter Bank Common Spatial Patterns (FBCSP) pipeline [214]. It first applies a temporal convolution to learn band-pass filters, followed by a spatial convolution to aggregate cross-channel structure. Its distinctive nonlinearity—elementwise squaring followed by a stabilized logarithm—produces variance-like features that mirror FBCSP's power-based representations. Table 12 reports the exact instantiation.

Table 12: ShallowConvNet architecture in symbolic form for 64-channel, 320-sample signals.

Layer name	Type	Input shape	Output shape	Param.#
Input	InputLayer	(64, 320, 1)	(1, 64, 320)	0
TemporalConv	Conv2D	(1, 64, 320)	(40, 64, 308)	560
SpatialConv	Conv2D	(40, 64, 308)	(40, 1, 308)	102,440
BN1	BatchNormalization	(40, 1, 308)	(40, 1, 308)	80
Activation1	Square	(40, 1, 308)	(40, 1, 308)	0
AvgPool1	AveragePooling2D	(40, 1, 308)	(40, 1, 40)	0
Activation2	SafeLog	(40, 1, 40)	(40, 1, 40)	0
Dropout1	Dropout	(40, 1, 40)	(40, 1, 40)	0
Flatten	Flatten	(40, 1, 40)	(1600)	0

For both backbones, the flattened feature vector(N_{feat}) is passed to a single fully connected layer with output dimension equal to the number of target classes C, followed by a softmax. This minimalist head avoids over-parameterization while preserving discriminative capacity (Table 13).

Table 13: Classifier head stacked on top of the backbone features.

Layer	Type	Input shape	Output shape	Param.#
Input	InputLayer	(N_{feat})	(N_{feat})	0
FC	Dense	(N_{feat})	(C)	$N_{feat} \times C$
Softmax	Activation	(C)	(C)	0

10.3 Training Details

The training methodology adheres to the standard protocol for unsupervised domain adaptation. In this paradigm, labeled data from a source domain are utilized concurrently with unlabeled data from a target domain. This approach aims to learn domain-invariant features without requiring supervisory signals from the target domain.

All models were trained using the Adam optimizer with a constant learning rate of $\eta_0 = 10^{-3}$ for both the EEGNet and ShallowConvNet backbones. To promote gradual domain alignment and mitigate early-stage overfitting, a dynamic scheduling scheme was applied exclusively to the adversarial loss weight. This schedule progressively increases the influence of the domain alignment term as training progresses.

The evaluation is conducted on the GIGA dataset using two distinct cross-validation schemes. The first, a **Cross-Session (intra-subject)** protocol, evaluates the model's ability to generalize across different recording sessions from the same individual. For each subject, a leave-one-session-out cross-validation is performed where one session is designated as the target domain for testing, and the remaining sessions from that subject

constitute the source domain for training. The second protocol, **Cross-Subject (inter-subject)**, assesses generalization across different individuals. This more challenging scenario employs a leave-one-subject-out cross-validation, where one subject's entire data is held out as the target domain, and the aggregated data from all other subjects form the source domain.

To prepare the data for each fold, the source and target datasets are partitioned independently. For the Cross-Session experiments, an 80% training and 20% testing split is used. For the more complex Cross-Subject task, the data is split into 70% for training, 15% for validation, and 15% for testing. A fixed random seed is used throughout all partitioning to ensure reproducibility. The batch size was set to 16 for the Cross-Session protocol and increased to 128 for the Cross-Subject protocol to accommodate the larger volume of data from multiple subjects.

The specific architectural configurations for EEGNet and ShallowConvNet are detailed in Tables 11 and 12, respectively. These parameters were held constant across all experiments. The models are implemented within the CREDA framework, which replaces the conventional adversarial loss with a divergence-based regularizer. This regularizer incorporates Rényi entropy to weight target samples based on their predictive confidence, enabling robust and class-consistent alignment without an explicit domain discriminator.

For all experiments, the kernel bandwidth parameter σ used in the estimation of Rényi's quadratic entropy was adaptively determined for each training batch using the median heuristic. This common practice involves setting σ as the square root of the median of all pairwise squared Euclidean distances within the combined source and target feature batch [212].

This data-driven approach automates a critical hyperparameter, ensuring that the kernel's scale is appropriately tailored to the feature distribution, which enhances the stability and effectiveness of the alignment process across domains.

To qualitatively assess the discriminative power of the learned representations, we employed Uniform Manifold Approximation and Projection (UMAP) for dimensionality reduction. Visualizations were generated using the GPU-accelerated `cuML` library, with default parameters. Prior to projection, features were normalized via a `MinMaxScaler` to facilitate visual inspection of inter-class and inter-domain separability.

All experiments were performed on cloud computing platforms (Google Colab, Kaggle). The development environment consisted of Python 3.11.11, with models implemented in PyTorch 2.1.2 and visualizations generated using `cuML` 25.02.01. The source code and datasets are publicly available at: `https://github.com/Daprosero/Domain_Adaptation` (accessed July 4, 2025).

10.4 Results and Discussion

10.4.1 Domain Adaptation Results

The empirical evaluation of CREDA on EEG data begins with an ablation study to ascertain the efficacy of its core components, particularly the noise-aware label weighting mechanism. Table 14 presents these results for both EEGNet and ShallowConvNet backbones under Cross-Session and Cross-Subject protocols. For the EEGNet architec-

ture, CREDA demonstrates a clear and consistent performance improvement over the Baseline across all metrics (Accuracy, F1-score, and Cohen's Kappa). Specifically, in the Cross-Session setting, CREDA improves the average accuracy by approximately 7.5 percentage points. The full CREDA model achieves a marginal but consistent gain over CREDA* (which lacks label weighting), suggesting that the entropy-based weighting scheme effectively contributes to stabilizing the training and refining the alignment, particularly in the more challenging Cross-Subject scenario. Conversely, for the ShallowConvNet backbone, the Baseline model shows highly competitive performance, even outperforming CREDA in the Cross-Session protocol. However, in the more complex Cross-Subject task, CREDA re-establishes a slight advantage, indicating that while ShallowConvNet may inherently learn more robust features for session-level shifts, CREDA's explicit domain adaptation is beneficial when faced with the greater variability across subjects.

Table 14: Ablation on EEG backbones, with the best method per block in **bold**, and CREDA* denoting our proposal without label weighting.

Backbone	**Protocol**	**Method**	**Acc. (%)**	**F1 (%)**	κ
EEGNet	Cross-Session	Baseline	60.70 ± 11.44	52.66 ± 13.70	0.217 ± 0.201
		CREDA*	68.19 ± 15.48	65.19 ± 17.62	0.355 ± 0.279
		CREDA	**68.24 ± 15.45**	**65.24 ± 17.60**	**0.356 ± 0.279**
	Cross-Subject	Baseline	63.97 ± 14.20	60.98 ± 16.85	0.282 ± 0.281
		CREDA*	66.94 ± 14.76	65.94 ± 15.78	0.339 ± 0.296
		CREDA	**67.08 ± 14.60**	**66.13 ± 15.51**	**0.342 ± 0.292**
ShallowConvNet	Cross-Session	Baseline	**66.66 ± 15.36**	**64.93 ± 16.73**	**0.313 ± 0.288**
		CREDA*	65.14 ± 15.28	63.82 ± 16.06	0.294 ± 0.303
		CREDA	65.02 ± 15.29	63.64 ± 16.09	0.292 ± 0.303
	Cross-Subject	Baseline	68.48 ± 13.53	65.97 ± 16.08	0.369 ± 0.272
		CREDA*	69.17 ± 14.55	**68.44 ± 15.15**	0.383 ± 0.293
		CREDA	**69.17 ± 14.47**	68.43 ± 15.07	**0.383 ± 0.291**

Figure 33 expands this analysis to include a comparison with established domain adaptation methods: DANN, ADDA, and CDAN+E. The plots illustrate the per-subject classification accuracy, highlighting the significant inter-subject variability inherent in BCI applications. No single method consistently outperforms all others across every subject. However, CREDA (represented by the green line) frequently positions itself within the top-performing cluster of models for a majority of subjects, particularly when paired with the EEGNet backbone. This underscores the challenge of developing a universally optimal approach and motivates a statistical aggregation of performance.

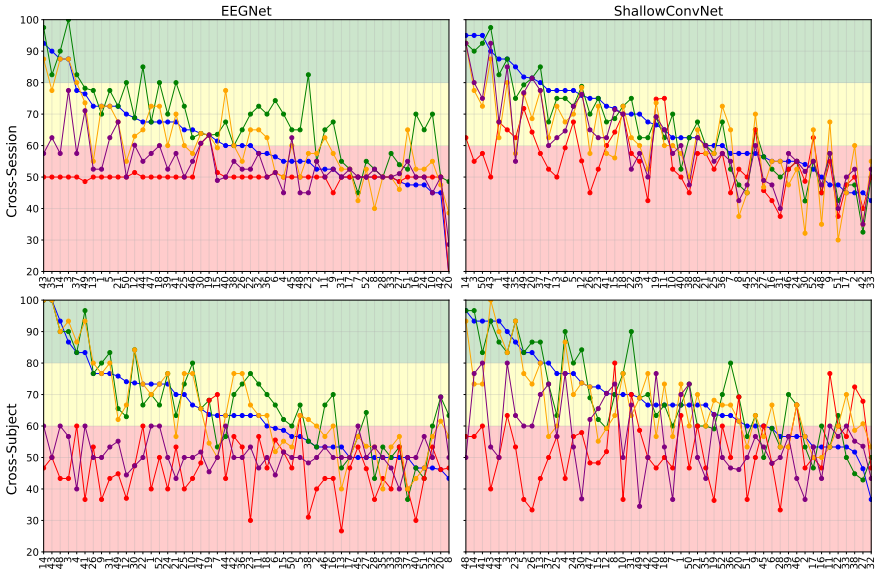


Figure 33: Cross-Session and Cross-Subject classification accuracy. Subjects are ordered according to the baseline performance of each backbone to ensure consistent model comparison.

To this end, Figure 34 and Table 15 provide a robust statistical assessment of the models' performance. The heatmaps on the left visualize the performance rank of each model for every subject, while the tables on the right display the p-values from pairwise t-tests, indicating the statistical significance of performance differences. Table 15 aggregates these findings, showing average ranks and p-values. In the Cross-Session protocol, EEGNet-CREDA emerges as the top-performing model with an average rank of 2.77, clearly distinguishing itself from the other methods. In the more demanding Cross-Subject protocol, the performance gap narrows, with ShallowConvNet-Baseline, ShallowConvNet-CREDA, and EEGNet-CREDA forming a competitive top tier. Notably, adversarial methods like ADDA and CDAN+E, despite their success in other fields, consistently rank lower, suggesting that their alignment mechanisms may be less suited to the subtle and complex nature of domain shifts in EEG data compared to CREDA's entropy-based regularization.

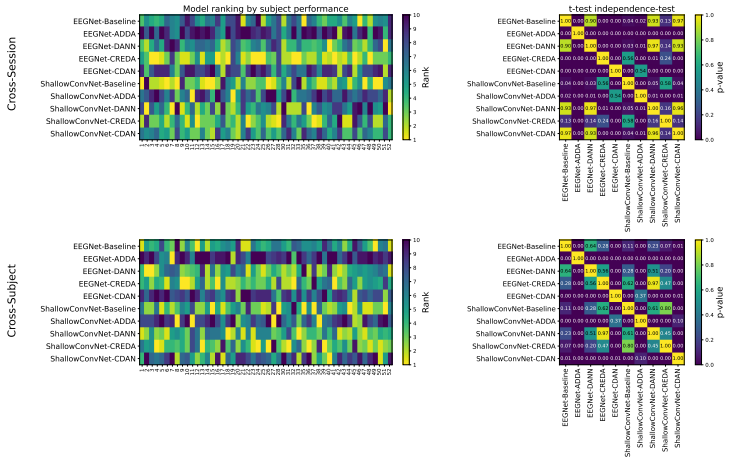


Figure 34: Average model rankings versus t-test p-values for Cross-Session and Cross-Subject scenarios.

Table 15: Average ranking and average t-test p-values for EEG models under cross-session and cross-subject protocols

Protocol	**Model**	**Avg. Ranking**	**Avg. t-test p-value**
Cross-Session	EEGNet-CREDA	2.77	0.09
	ShallowConvNet-Baseline	3.54	0.14
	ShallowConvNet-CREDA	4.17	0.16
	EEGNet-DANN	5.18	0.33
	ShallowConvNet-CDAN	5.43	0.34
	ShallowConvNet-DANN	5.51	0.34
	EEGNet-Baseline	5.58	0.33
	ShallowConvNet-ADDA	7.16	0.07
	EEGNet-CDAN	7.31	0.06
	EEGNet-ADDA	8.35	0.00
Cross-Subject	ShallowConvNet-Baseline	3.85	0.27
	ShallowConvNet-CREDA	3.91	0.22
	EEGNet-CREDA	4.02	0.32
	ShallowConvNet-DANN	4.05	0.31
	EEGNet-DANN	4.42	0.24
	EEGNet-Baseline	5.01	0.15
	ShallowConvNet-CDAN	6.51	0.01
	ShallowConvNet-ADDA	7.14	0.05
	EEGNet-CDAN	7.64	0.04
	EEGNet-ADDA	8.45	0.00

Figure 35 offers a more nuanced perspective by stratifying subjects into high, medium, and low-performance groups. For high-performing subjects (accuracy ¿ 80%),

most models perform well, and the margin for improvement is small. The benefits of domain adaptation become most apparent in the medium-performance group (60%-80% accuracy), where methods like CREDA and DANN often provide a substantial boost over the baseline. For low-performing subjects ($\leq 60\%$), a phenomenon often associated with BCI illiteracy, all models struggle, indicating that the underlying signal may lack sufficient class-discriminative information for any algorithm to leverage effectively. This analysis suggests that CREDA is particularly effective for subjects for whom a usable BCI is challenging but achievable, representing a critical user group for practical applications.

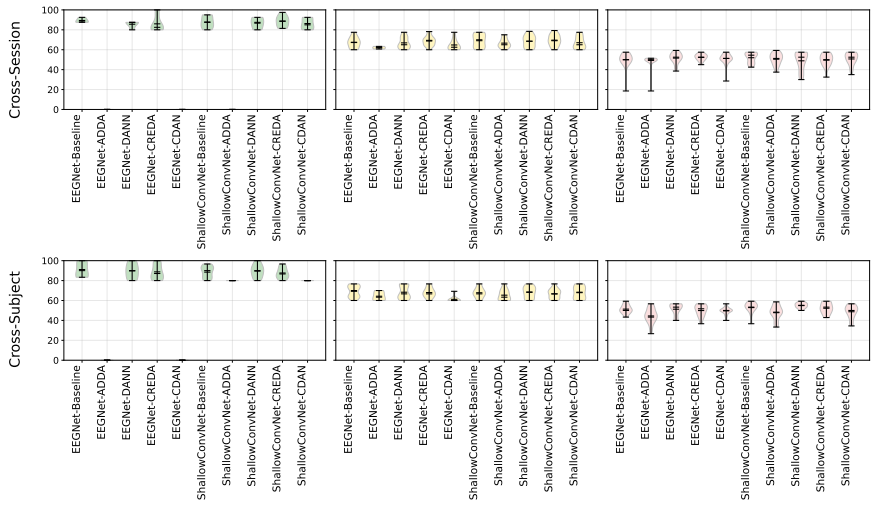


Figure 35: Cross-Session and Cross-Subject group-performing MI-EEG classification results. **Left:** Subjects with model accuracy above 80%. **Middle:** Subjects with model accuracy between 60% and 80%. **Right:** Subjects with model accuracy below 60%.

10.4.2 Interpretability Results

To qualitatively understand the mechanisms behind the observed performance differences, we visualized the learned feature representations using Uniform Manifold Approximation and Projection (UMAP). Figure 36 depicts the two-dimensional projection of the original EEG features, prior to any adaptation. The visualization reveals a dense, unstructured cloud of data points where different sessions are completely entangled. This illustrates the fundamental challenge: the absence of a clear separation between domains in the initial feature space, which complicates the task of learning domain-invariant yet class-discriminative representations.

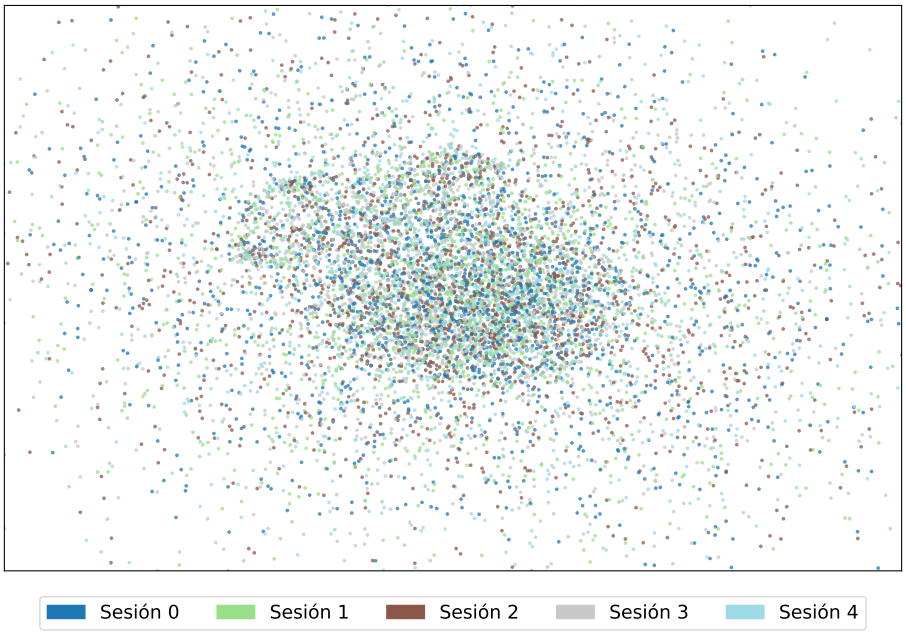


Figure 36: Two-dimensional UMAP projections of original EEG feature representations before domain adaptation.

Figure 37 demonstrates how different adaptation methods address this challenge in the Cross-Session scenario. The rows correspond to different subjects, and columns represent the applied method. The Baseline model fails to impose any significant structure, leaving the source (circle) and target (triangle) domains mixed and the classes poorly separated. While adversarial methods like DANN and ADDA achieve some degree of domain alignment, it often comes at the cost of class coherence, resulting in fragmented and overlapping class clusters. CREDA, in stark contrast, consistently yields a superior latent space configuration. It successfully aligns the source and target domains—evidenced by the homogeneous mixture of circles and triangles—while simultaneously enhancing the separability between Class 0 (blue) and Class 1 (orange). This visual evidence directly corroborates the objective of CREDA's class-conditional alignment loss, which is designed to minimize divergence between domains on a per-class basis, thereby preserving the semantic structure essential for accurate classification.

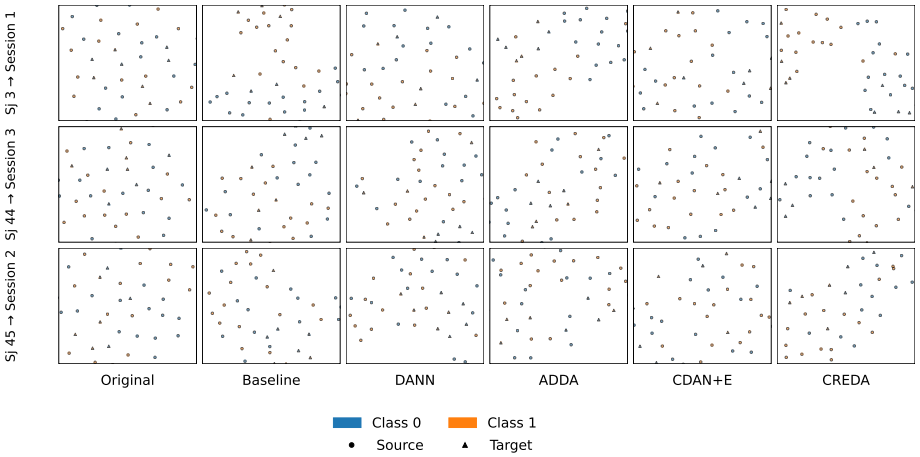


Figure 37: Cross-Session UMAP projections of learned feature representations across domain adaptation methods. Points are colored by class, and marker shape encodes domain. **Rows**: evaluation subject and session. **Columns**: compared adaptation models.

Finally, Figure 38 visualizes the feature space learned by CREDA in the even more challenging Cross-Subject setting. Each panel shows the alignment of a single target subject's data (e.g., Subject 14) against data from multiple source subjects (e.g., Subjects 43, 41, 31, distinguished by marker shape). The projections demonstrate CREDA's ability to construct a unified feature space that transcends individual neurophysiological differences. Despite the inherent variability across subjects, CREDA maps their corresponding neural patterns into a shared manifold where the fundamental class structure (blue vs. red points) remains distinct and separable. This powerful capability for generalization highlights the potential of CREDA to build more robust and user-independent BCI systems, mitigating the need for extensive subject-specific calibration.

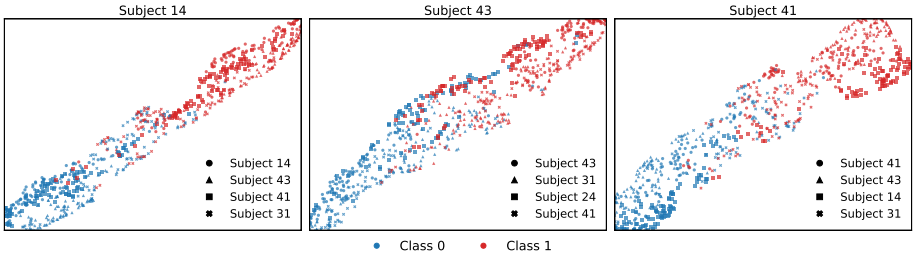


Figure 38: Cross-Subject UMAP projections of learned EEG feature representations under CREDA. Points are colored by class. The evaluation (target) subject is shown alongside multiple source subjects, which are distinguished by marker shapes.

11 Limitations

Despite the robust performance of the CREDA framework, several limitations—many of which were highlighted during the review process—must be acknowledged. These, in turn, outline compelling directions for future research.

Firstly, as noted by the committee, while CREDA demonstrates superior performance on deep CNNs, its validation on EEG data was confined to moderate-scale models (e.g., EEGNet). The results revealed an architectural dependency, which underscores the need for a comprehensive investigation to fully characterize its scaling properties with more advanced, state-of-the-art EEG architectures.

Secondly, the current implementation employs a singular, globally-tuned hyperparameter strategy, precluding domain-pair-specific optimization. This includes a static weighting for loss functions, a fixed Rényi entropy order of $\alpha = 2$, and a non-optimized kernel bandwidth. While the choice of $\alpha = 2$ is justified for its stable matrix form, exploring its role as a tunable hyperparameter could foster more stable and adaptive training dynamics, a notable challenge given the high variability inherent in EEG signals.

Thirdly, a key limitation emerges in scenarios with extreme domain shifts. As discussed, this is attributed to the generation of systematically corrupted pseudo-labels, which undermines the class-conditional alignment mechanism. Although our EEG experiments focused on Cross-Session and Cross-Subject adaptation, this finding strongly suggests that CREDA may face similar challenges in more demanding **Cross-Dataset** scenarios, requiring new strategies to mitigate the impact of corrupted labels.

12 Conclusions

This work introduced a novel domain adaptation framework, termed Conditional Rényi α-Entropy Domain Adaptation (CREDA), a deep learning-based strategy integrating kernel-based conditional alignment from a matrix-based formulation of Rényi's quadratic entropy. CREDA is structured around three key components: a deep feature extractor for learning domain-invariant representations, an entropy-weighted strategy to enhance robustness against noisy pseudo-labels, and a class-conditional alignment loss to promote semantic consistency across domains.

Experimental results across diverse visual and neurophysiological adaptation scenarios demonstrate that CREDA consistently outperforms conventional methods such as DANN, ADDA, and CDAN+E in terms of predictive accuracy, representational quality, and interpretability. For visual tasks, CREDA achieves the highest average accuracy across all datasets and architectures, showing remarkable robustness on deeper CNNs (ResNet-50) and attention-based models (ViT-Tiny) where adversarial approaches often degrade. In the challenging EEG domain, CREDA again demonstrated top-tier performance, particularly in the complex Cross-Subject protocol, effectively mitigating the high inter-subject variability that plagues BCI systems.

Notably, CREDA consistently maintains class separability under complex distribution shifts, a finding corroborated by UMAP visualizations for both image and EEG data, which revealed a superiorly organized latent space compared to competing methods. The integration of Grad-CAM++ based visualizations further reinforces its applicability

in real-world settings where traceability and semantic coherence are critical. From an implementation standpoint, CREDA's modular architecture, confidence-aware weighting, and class-conditional regularization enhance robustness and facilitate seamless integration into existing deep learning pipelines, making it a powerful and versatile tool for transfer learning.

13 Future Work

Building on these results and the valuable feedback received, our future work will be directed towards addressing the identified limitations and expanding the framework's capabilities.

A primary theoretical direction will be to formalize the relationship between the alignment term and a theoretical performance bound, as suggested by the committee. Concurrently, to address the $O(n^2)$ complexity of the Gram matrix computation, we will investigate and implement low-rank approximation methods, such as the Nyström method and Random Fourier Features (RFF), to accelerate training and inference.

Our research will also extend CREDA to more complex adaptation settings, including multi-source, continual, and source-free domain adaptation, which are highly pertinent to real-world BCI applications. This aligns with the recommendation to benchmark against more contemporary baselines from 2023-2025.

Furthermore, we will significantly enhance the framework's interpretability. In addition to our current use of UMAP and Grad-CAM++, we will incorporate neurophysiologically-grounded visualizations, such as topomaps and spectro-temporal analyses, to provide deeper insights specific to EEG data. To further reinforce the validity of our explanations, we plan to explore the integration of causal inference principles, addressing the need for a more robust and explicit explanatory mechanism.

Finally, our long-term vision is to leverage these advancements to construct a foundational model for EEG domain generalization. This involves extending CREDA to challenging *few-shot* scenarios and pre-training it on large-scale datasets. The ultimate objective is to engineer a single, robust, and computationally efficient model that can be rapidly adapted to new users with minimal data, supported by a fully reproducible suite of artifacts, including fixed seeds, configuration files, and scripts, to ensure the transparency and reliability of our findings.

14 Academic products

- Pérez-Rosero, D.A.; Manrique-Cabezas, D.A.; Triana-Martinez, J.C.; Álvarez-Meza, A.M.; Castellanos-Dominguez, G. An Explainable Framework Integrating Local Biplots and Gaussian Processes for Unemployment Rate Prediction in Colombia. Computation 2025, 13, 116. `https://doi.org/10.3390/computation13050116`. Q2

- Pérez-Rosero, D.A.; Álvarez-Meza, A.M.; Castellanos-Domínguez, G. *Conditional Domain Adaptation with α-Rényi Entropy Regularization and Noise-Aware*

Label Weighting. Mathematics 2025, 13, 2602. https://doi.org/10.3390/math13162602.

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